香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

ELEC 3200: System Modeling, Analysis and Control

Lecture 20-22:
Root Locus Method

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Course structure so far:

$$
\begin{array}{rcl}
\text{modeling} & — & \text{examples} \\
\downarrow & & \\
\text{analysis} & — & \text{transfer function, response, stability} \\
\downarrow & & \\
\text{design} & — & \text{some simple examples given}
\end{array}
$$

We have known that the performance of a system is largely determined by the locations of the system’s poles.

- If all system parameters are known, the pole locations can be easily determined using model or computer, such as those demonstrated in our previous lectures.
- However, in real practice, it is often more important to understand how the closed-loop poles vary when certain system parameters are changed.

How the poles vary when certain system parameters are changed?

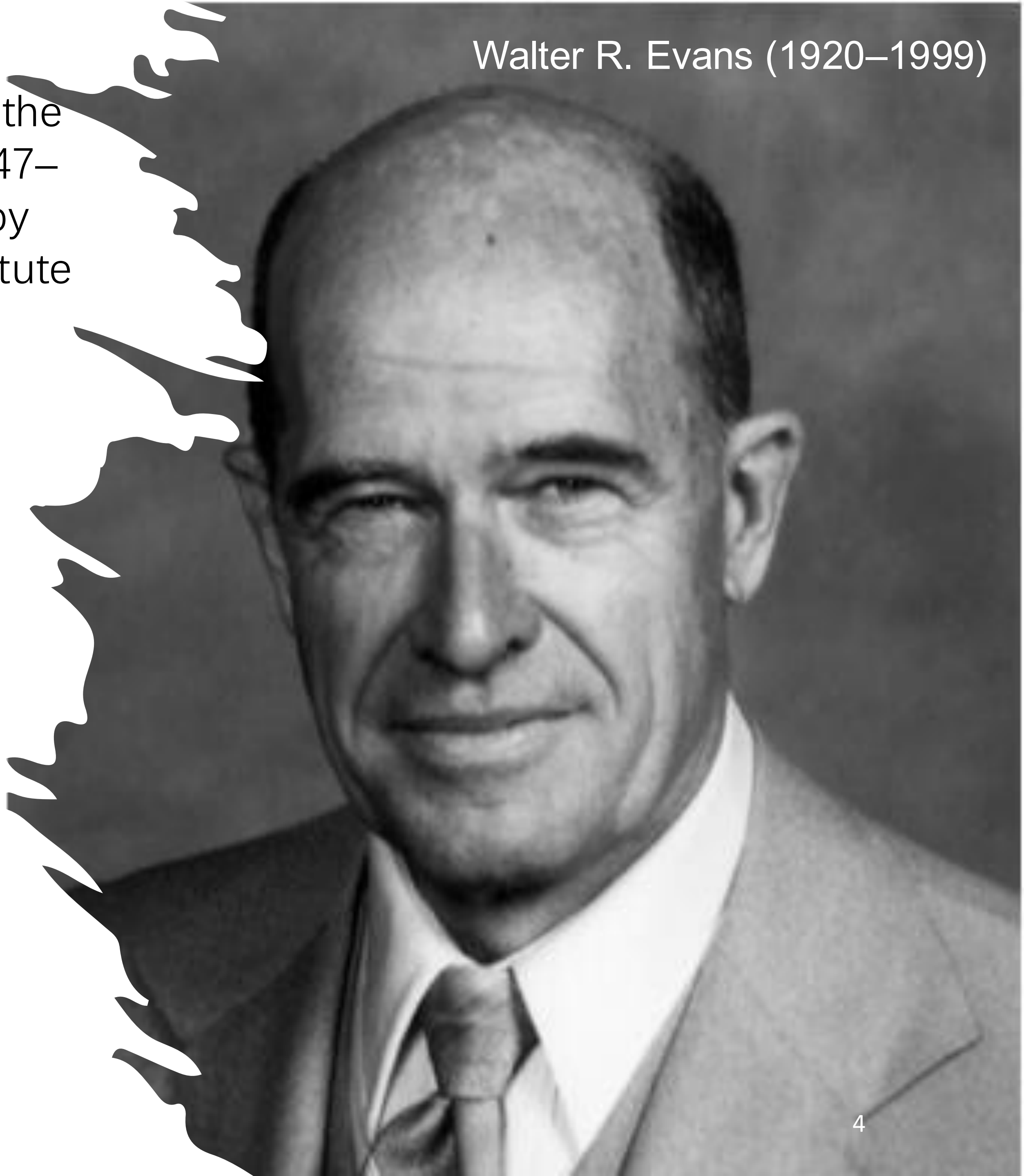
The reasons we are interested in this kind of problem:

- Many parameters in a system may vary with the working environment or cannot be determined exactly. It is then important to ensure that the system can still operate in the desired state.
- Control engineers need to know how certain controller parameters affect system performance such that the desired controller parameters can be chosen to achieve the design objectives.

One classical technique in determining pole variations with parameters is known as the root-locus method, invented by ***W. R. Evens***.

Evans' root-locus method was published in the paper "Graphical analysis of control systems," Transactions of the American Institute of Electrical Engineers, vol. 67, pp. 547–551, 1948, and in the paper "Control system synthesis by root-locus method," Transactions of the American Institute of Electrical Engineers, vol. 69, pp. 66–69, 1950.

- It should be pointed out that in most cases it is more important to know how to quickly sketch a root locus, which would indicate the trend of the root loci rather than the exact root-locus plot, which can be done by using a computer if necessary.
- Thus, attention is not focused on the exact details of root-locus construction in this lecture. Instead, we shall concentrate on how we can use root locus as a tool for system analysis and controller design.



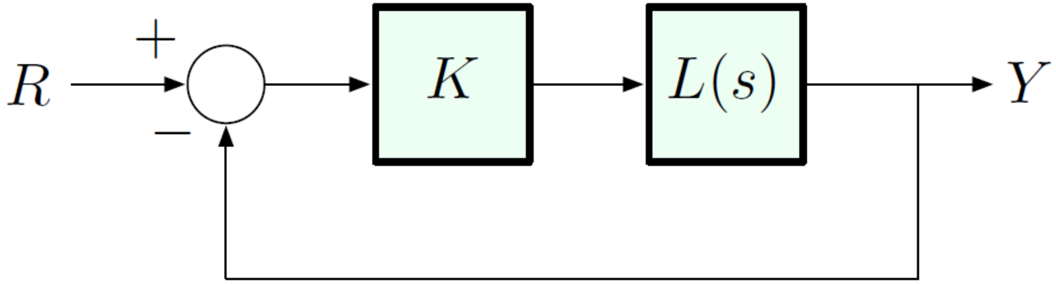
Walter R. Evans (1920–1999)

The Root Locus Design Method

(invented by Walter R. Evans in 1948)

$$\frac{Y}{R} = \frac{K\frac{b}{a}}{1+K\frac{b}{a}} = \frac{Kb}{a+Kb}$$

Consider this unity feedback configuration:

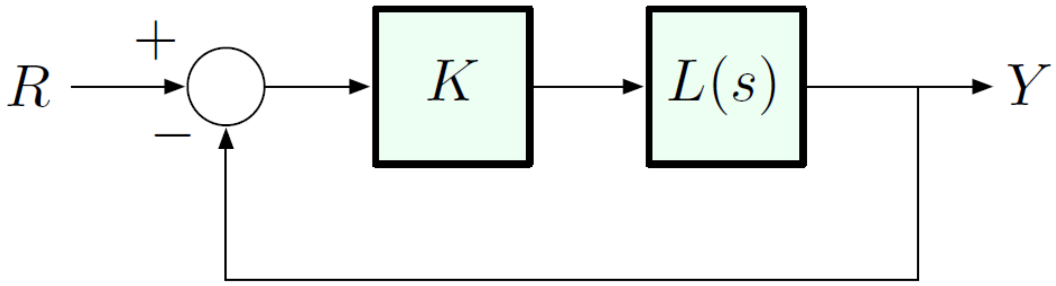


$$C(s) = a + Kb$$

where

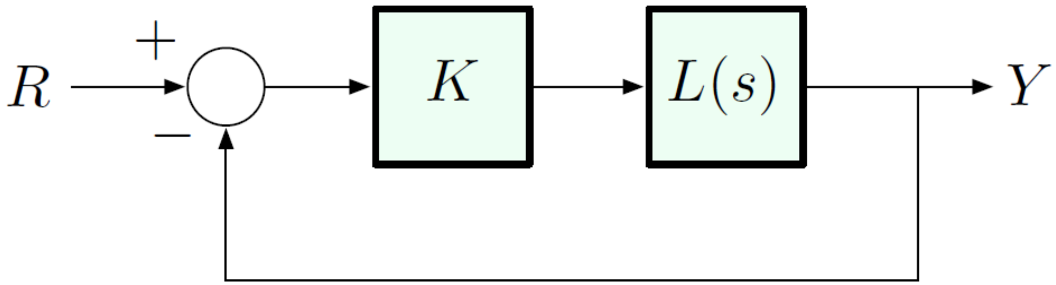
- K is a constant gain
- $L(s) = \dfrac{b(s)}{a(s)}$, where $a(s)$ and $b(s)$ are some polynomials

The Root Locus Design Method



Closed-loop transfer function: $\dfrac{Y}{R} = \dfrac{KL(s)}{1 + KL(s)}, \; L(s) = \dfrac{b(s)}{a(s)}$

The Root Locus Design Method



Closed-loop transfer function: $\dfrac{Y}{R} = \dfrac{KL(s)}{1 + KL(s)}$, $L(s) = \dfrac{b(s)}{a(s)}$

Closed loop poles are solutions of:

$$1 + KL(s) = 0 \qquad \Leftrightarrow \qquad L(s) = -\frac{1}{K}$$

$$\Updownarrow$$

$$1 + \frac{Kb(s)}{a(s)} = 0$$

$$\Updownarrow$$

$$\underbrace{a(s) + Kb(s)}_{\substack{\text{characteristic}\\ \text{polynomial}}} = 0 \qquad \text{characteristic equation}$$

A Comment on Change of Notation

Note the change of notation:

$$\text{from } H(s) \text{ or } G(s) = \frac{q(s)}{p(s)} \quad \text{to } L(s) = \frac{b(s)}{a(s)}$$

— the RL method is quite general, so $L(s)$ is not necessarily the *plant* transfer function, and K is not necessary *feedback gain* (could be *any parameter*).

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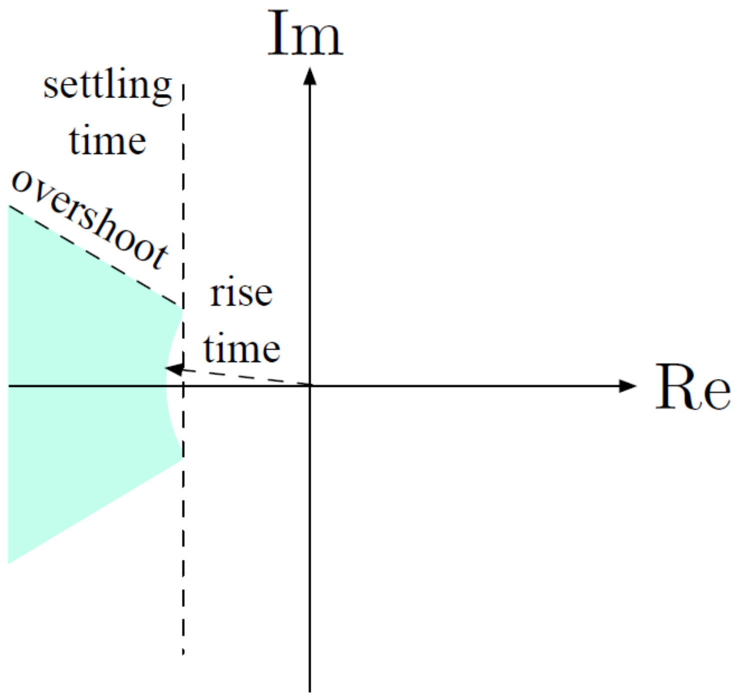
E.g., $L(s)$ and K may be related to plant transfer function and feedback gain through some transformation.

As long as we can represent the poles of the closed-loop transfer function as roots of the equation $1 + KL(s) = 0$ for *some choice* of K and $L(s)$, we can apply the RL method.

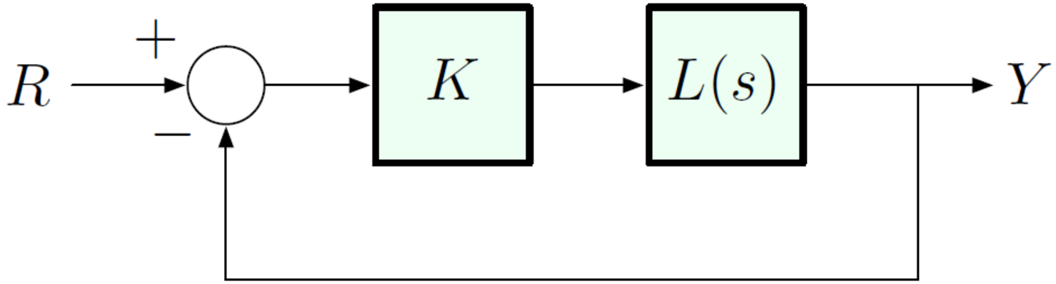
Towards Quantitative Characterization of Stability

Qualitative description of stability: Routh test gives us a range of K to guarantee stability.



For what values of K do we best satisfy given design specs?

Root Locus and Quantitative Stability



Closed-loop transfer function: $\dfrac{Y}{R} = \dfrac{KL(s)}{1+KL(s)}, \quad L(s) = \dfrac{b(s)}{a(s)}$

For what values of K do we best satisfy given design specs?

Specs are encoded in pole locations, so:

The *root locus* for $1 + KL(s)$ is the set of all closed-loop poles, i.e., the roots of

$$1 + KL(s) = 0,$$

as K varies from 0 to ∞. equivalent formula

A Simple Example

$$L(s) = \frac{1}{s^2 + s} \qquad b(s) = 1,\ a(s) = s^2 + s$$

Characteristic equation: ☆ $a(s) + Kb(s) = 0$

$$s^2 + s + K = 0$$

$$s = \frac{-1 \pm \sqrt{1-4K}}{2}$$

A Simple Example

$$L(s) = \frac{1}{s^2 + s} \qquad b(s) = 1,\ a(s) = s^2 + s$$

Characteristic equation: $\qquad a(s) + Kb(s) = 0$

$$s^2 + s + K = 0$$

Here, we can just use the quadratic formula:

$$s = -\frac{1 \pm \sqrt{1-4K}}{2} = -\frac{1}{2} \pm \frac{\sqrt{1-4K}}{2}$$

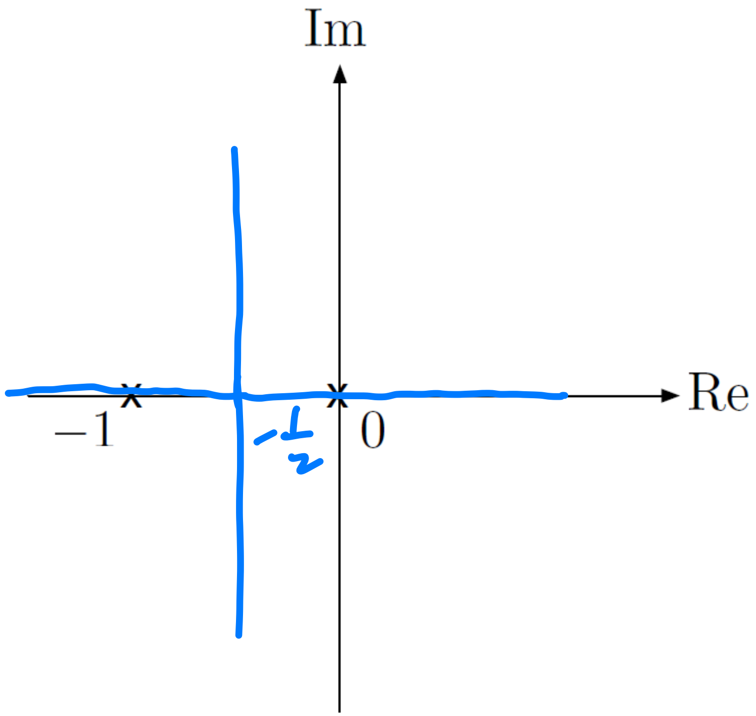
$$\text{Root locus} = \left\{ -\frac{1}{2} \pm \frac{\sqrt{1-4K}}{2} \ : 0 \le K < \infty \right\} \subset \mathbb{C}$$

Example, continued

$$\text{Root locus} = \left\{ -\frac{1}{2} \pm \frac{\sqrt{1-4K}}{2} \; : 0 \le K < \infty \right\} \subset \mathbb{C}$$

Let's plot it in the s-plane:

- start at $K = 0$ the roots are $-\frac{1}{2} \pm \frac{1}{2} \equiv -1, 0$
  note: these are poles of L (open-loop poles)

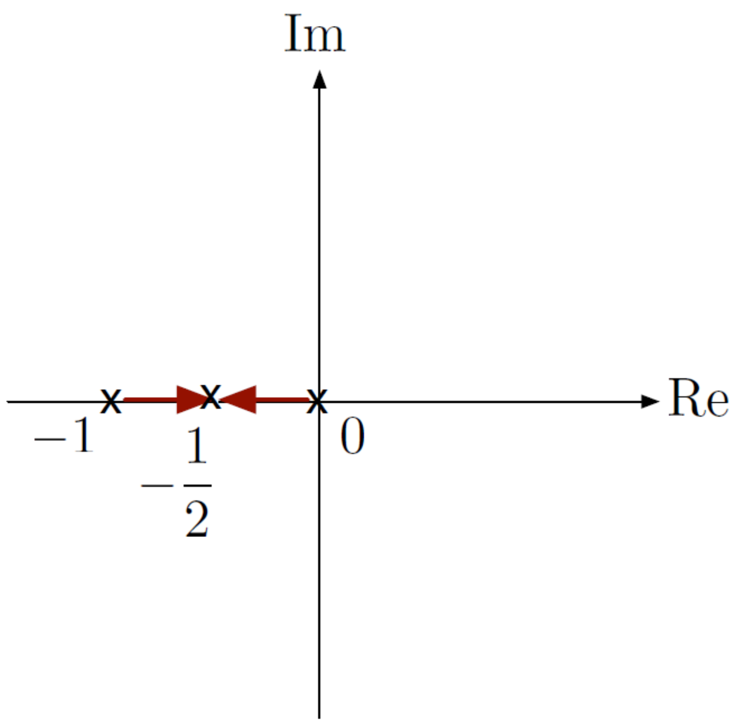


Example, continued

Root locus: $\left\{-\frac{1}{2} \pm \frac{\sqrt{1-4K}}{2} \;:\; 0 \le K < \infty\right\} \subset \mathbb{C}$

- as K increases from 0, the poles start to move

$$\begin{aligned} 1 - 4K > 0 \quad &\Longrightarrow \; 2 \text{ real roots} \\ K = 1/4 \quad &\Longrightarrow 1 \text{ real root } s = -1/2 \end{aligned}$$

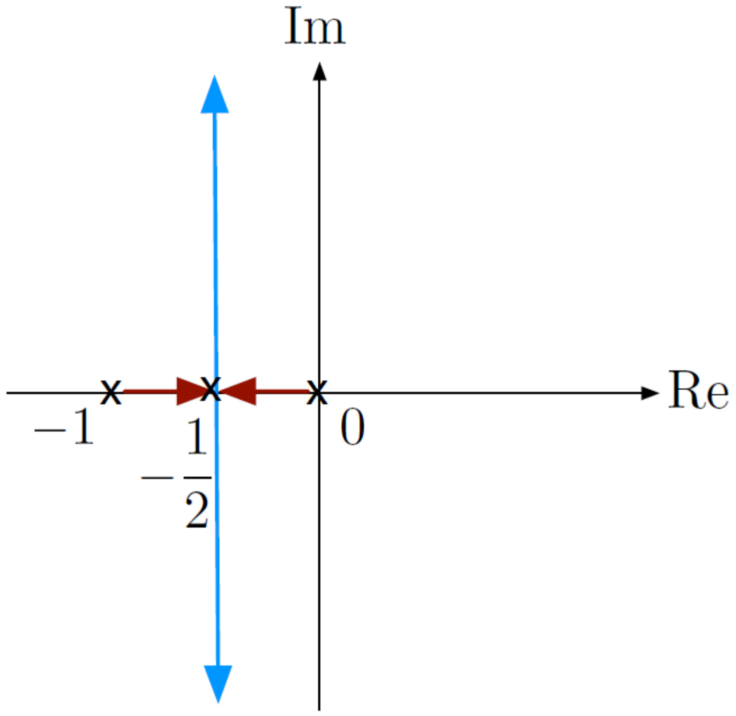


Example, continued

$$\text{Root locus: } \left\{ -\frac{1}{2} \pm \frac{\sqrt{1-4K}}{2} \ : 0 \le K < \infty \right\} \subset \mathbb{C}$$

- as K increases from 0, the poles start to move

$$K > 1/4 \qquad \Longrightarrow \quad 2 \text{ complex roots with } \mathrm{Re}(s) = -1/2$$

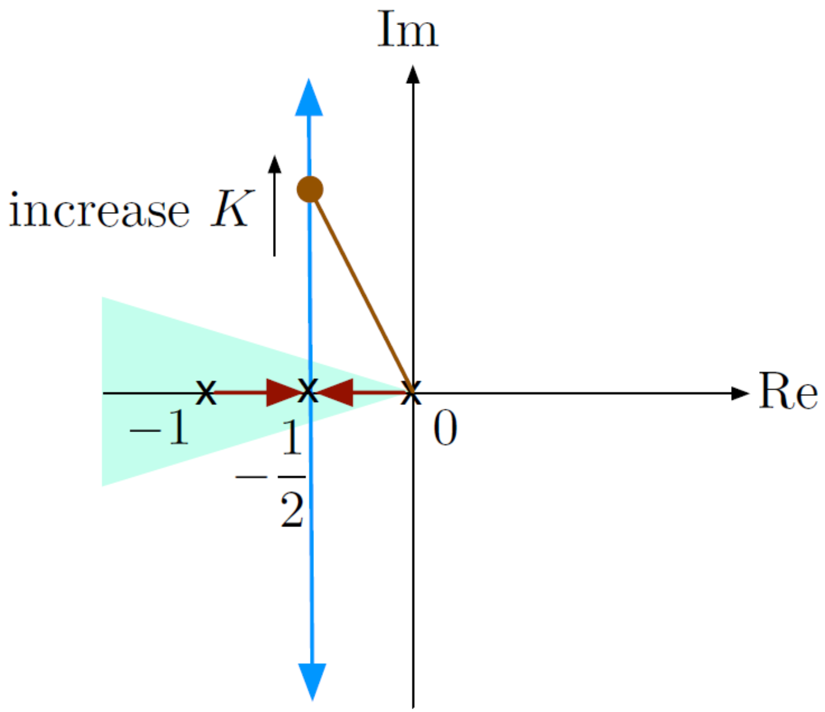


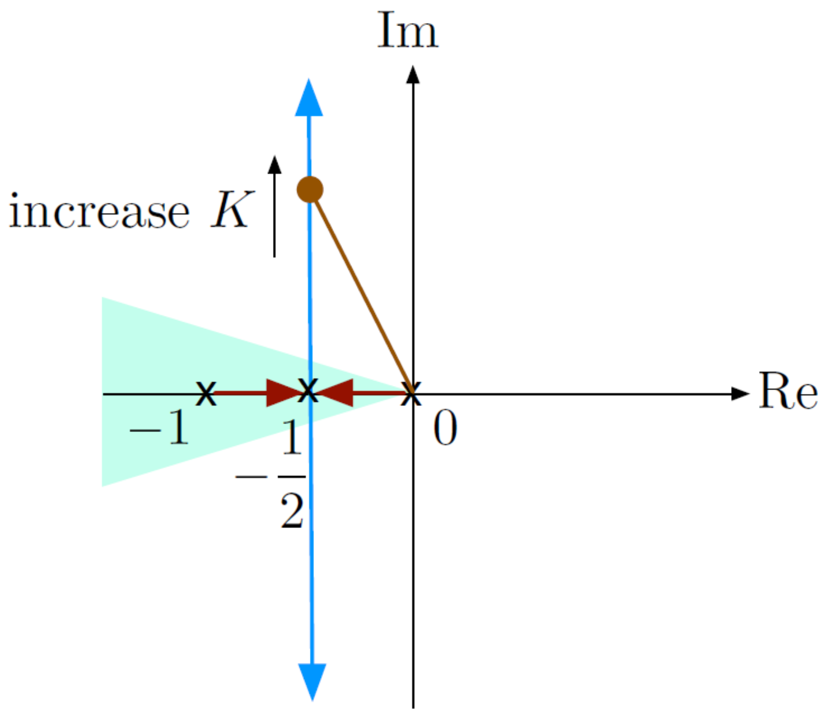
($s = -1/2$ is the *point of breakaway* from the real axis)

Example, continued

Compare this to admissible regions for given specs:

$t_s \approx \dfrac{3}{\sigma}$	want σ large, can only have $\sigma = \dfrac{1}{2}$ $(t_s = 6)$
$t_r \approx \dfrac{1.8}{\omega_n}$	want ω_n large $\Longrightarrow$ want K large
M_p	want to be inside the shaded region $\Longrightarrow$ want K small





Thus, the root locus helps us *visualize the trade-off* between all the specs in terms of K.

However, for order > 2, there will generally be no direct formula for the closed-loop poles as a function of K.

Our goal: develop simple rules for (approximately) sketching the root locus in the general case.

Equivalent Characterization of RL: Phase Condition

Recall our original definition: The *root locus* for $1 + KL(s)$ is the set of all closed-loop poles, i.e., the roots of

$$1 + KL(s) = 0,$$

as K varies from 0 to ∞.

A point $s \in \mathbb{C}$ is on the RL if and only if

$$L(s) = \underbrace{-\frac{1}{K}}_{\text{negative and real}} \quad \text{for some } K > 0$$

This gives us an equivalent characterization:

The phase condition: The root locus of $1 + KL(s)$ is the set of all $s \in \mathbb{C}$, such that $\angle L(s) = 180^\circ$, i.e., $L(s)$ is real and negative.

Six Rules for Sketching Root Loci

There are *six rules* for sketching root loci. These rules are mainly qualitative, and their purpose is to give intuition about impact of poles and zeros on performance.

These rules are:

- Rule A — number of branches
- Rule B — start points
- Rule C — end points
- Rule D — real locus
- Rule E — asymptotes
- Rule F — $j\omega$-crossings

Rule A: Number of Branches

$$1 + K\frac{b(s)}{a(s)} = 1 + K\frac{s^m + b_1 s^{m-1} + \ldots + b_{m-1}s + b_m}{s^n + a_1 s^{n-1} + \ldots + a_{n-1}s + a_n} = 0$$

Rule A: Number of Branches

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$$\implies (s^n + a_1 s^{n-1} + \ldots + a_{n-1}s + a_n) + K(s^m + b_1 s^{m-1} + \ldots + b_{m-1}s + b_m) = 0$$

Since $\deg(a) = n \geq m = \deg(b)$, the characteristic polynomial $a(s) + Kb(s) = 0$ has degree n.

The characteristic polynomial has n solutions (roots), some of which may be repeated. As we vary K, these n solutions also vary to form n branches.

Rule A:

$$\#(\text{branches}) = \deg(a)$$

Rule B: Start Points

The locus starts from $K = 0$. What happens near $K = 0$?

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If $a(s) + Kb(s) = 0$ and $K \sim 0$, then $a(s) \approx 0$.

Therefore:

- s is close to a root of $a(s) = 0$, or
- s is close to a pole of $L(s)$

Rule B: branches start at open-loop poles.

Rule C: End Points

What happens to the locus as $K \to \infty$?

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What happens to the locus as $K \to \infty$?

$$a(s) + Kb(s) = 0$$
$$b(s) = -\frac{1}{K}a(s)$$

— as $K \to \infty$,

- branches end at the roots of $b(s) = 0$, or
- branches end at zeros of $L(s)$

Rule C: branches end at open-loop zeros.

Rule C: End Points

What happens to the locus as $K \to \infty$?

$$a(s) + Kb(s) = 0$$
$$b(s) = -\frac{1}{K}a(s)$$

— as $K \to \infty$,

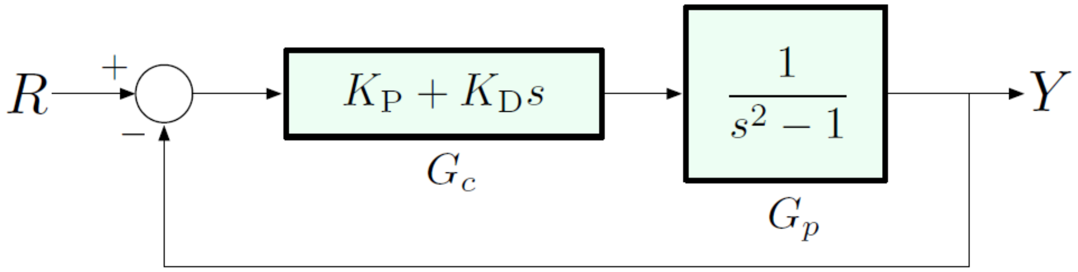
- branches end at the roots of $b(s) = 0$, or
- branches end at zeros of $L(s)$

Rule C: branches end at open-loop zeros.

Note: if $n > m$, we have n branches, but only m zeros. The remaining $n - m$ branches go off to infinity (end at "zeros at infinity").

Example

PD control of an unstable 2nd-order plant



Example

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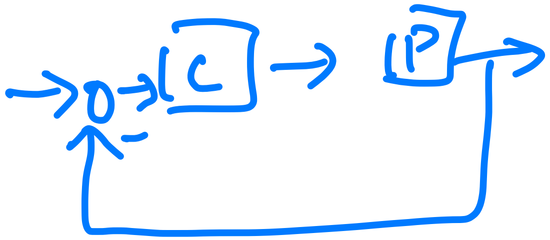
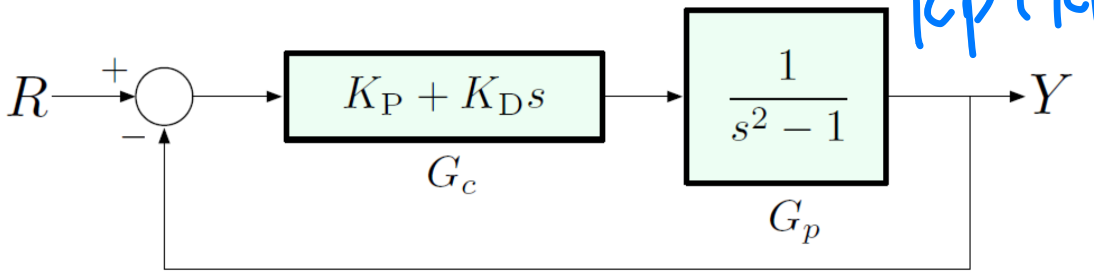


$$KL(s) = (k_P + k_D s)\frac{1}{s^2-1}$$



$$= K_D \frac{\left(\frac{k_P}{k_D} + s\right)}{s^2-1}$$

$$\frac{Y}{R} = \frac{G_c G_p}{1 + G_c G_p} \qquad \text{poles: } 1 + G_c(s)G_p(s) = 0$$

highest order

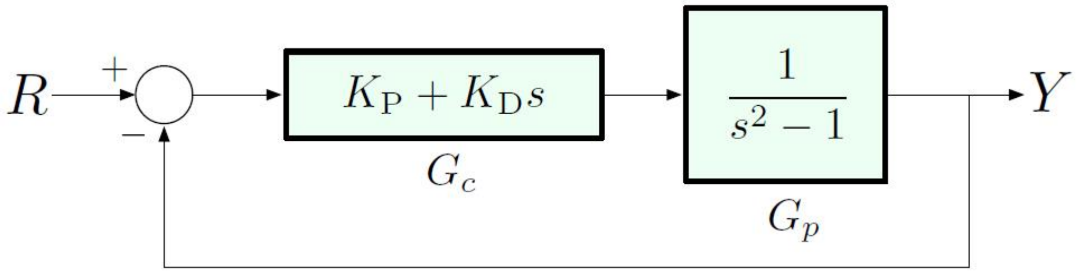
$$1 + (K_P + K_D s)\left(\frac{1}{s^2-1}\right) = 0$$

We will examine the impact of varying $K = K_D$, assuming the ratio K_P/K_D *fixed.*

$$C(s) = 1 + KL = s^2 - 1 + (k_P + k_D s) = 0$$

Example

PD control of an unstable 2nd-order plant



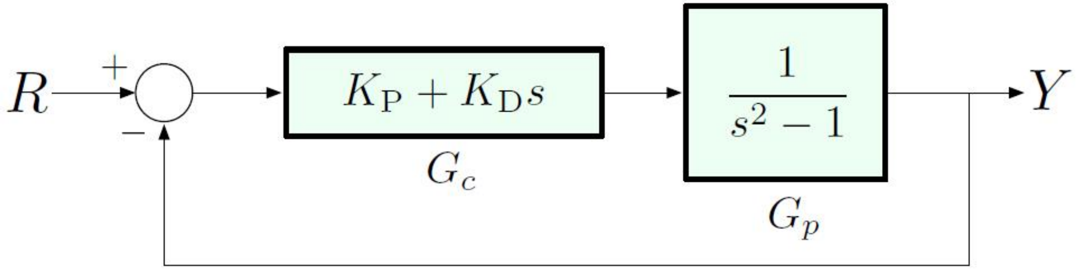
We will examine the impact of varying $K = K_D$, assuming the ratio K_P/K_D *fixed.*

Let us write the characteristic equation in *Evans form*:

$$1 + \underbrace{K_D}_{K}\left(s + \frac{K_P}{K_D}\right)\left(\frac{1}{s^2-1}\right) = 1 + K\underbrace{\frac{s + K_P/K_D}{s^2-1}}_{L(s)} = 0$$

Example

PD control of an unstable 2nd-order plant



We will examine the impact of varying $K = K_\mathrm{D}$, assuming the ratio $K_\mathrm{P}/K_\mathrm{D}$ *fixed*.

Let us write the characteristic equation in *Evans form*:

$s = \frac{-K_P}{K_D}$

$$1 + \underbrace{K_\mathrm{D}}_{K} \left(s + \frac{K_\mathrm{P}}{K_\mathrm{D}} \right) \left(\frac{1}{s^2 - 1} \right) = 1 + K \underbrace{\frac{s + K_\mathrm{P}/K_\mathrm{D}}{s^2 - 1}}_{L(s)} = 0$$

$s = \pm 1$

2 branches

$$L(s) = \frac{s - z_1}{s^2 - 1} \qquad \text{zero at } s = z_1 = -K_\mathrm{P}/K_\mathrm{D} < 0$$

Example

$$L(s) = \frac{s - z_1}{s^2 - 1}$$

- Rule A: $\begin{cases} m = 1 \\ n = 2 \end{cases} \implies$ 2 branches

Example

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Example

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- Rule B: branches start at open-loop poles $\quad s = \pm 1$
- Rule C: branches end at open-loop zeros $\quad s = z_1, -\infty$

Let $z_1 = -\frac{K_P}{K_D}$

why $-\infty$?
Will see it later!

Example

$$L(s) = \frac{s - z_1}{s^2 - 1}$$

$s^2 - 1 = 0$

$s = \pm 1$

- Rule A: $\begin{cases} m = 1 \\ n = 2 \end{cases} \implies$ 2 branches
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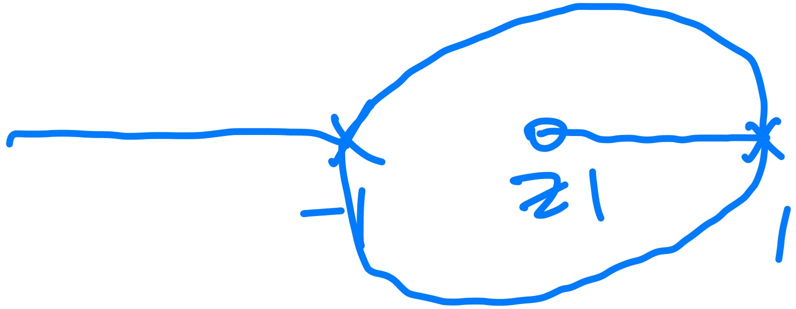


⇒ Design to add pole/zero by controller

Example

$$L(s) = \frac{s - z_1}{s^2 - 1}$$

- Rule A: $\begin{cases} m = 1 \\ n = 2 \end{cases} \implies$ 2 branches
- Rule B: branches start at open-loop poles $\qquad s = \pm 1$
- Rule C: branches end at open-loop zeros $\qquad s = z_1, -\infty$
  (we will see why $-\infty$ later)

So the root locus will look something like this:

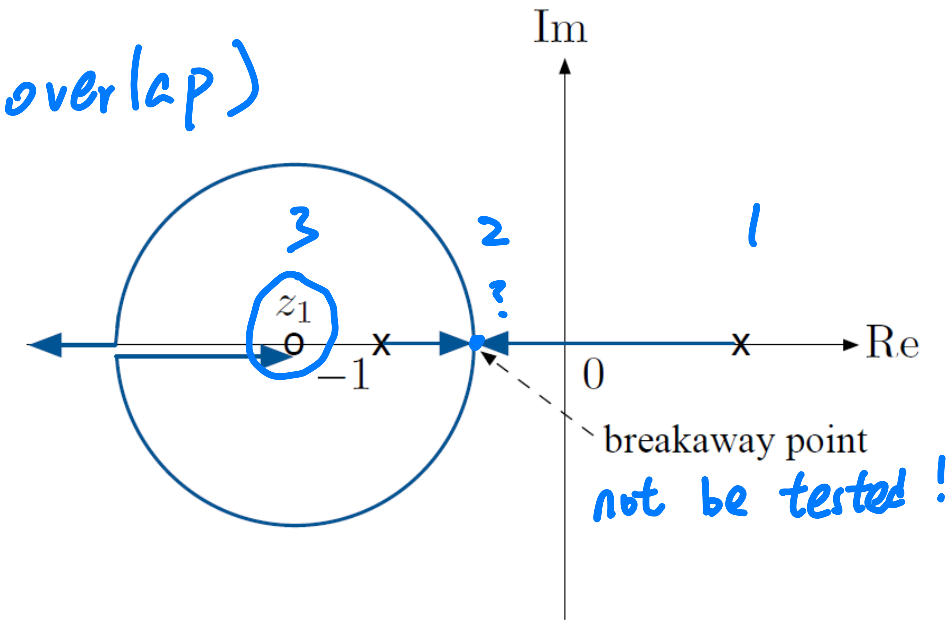
Not in exam!

$$\frac{dL}{ds} = 0 \quad \text{(overlap)}$$

$$(s+p)^2 = 0$$

$$2(s+p) = 0$$

$$s = -p$$

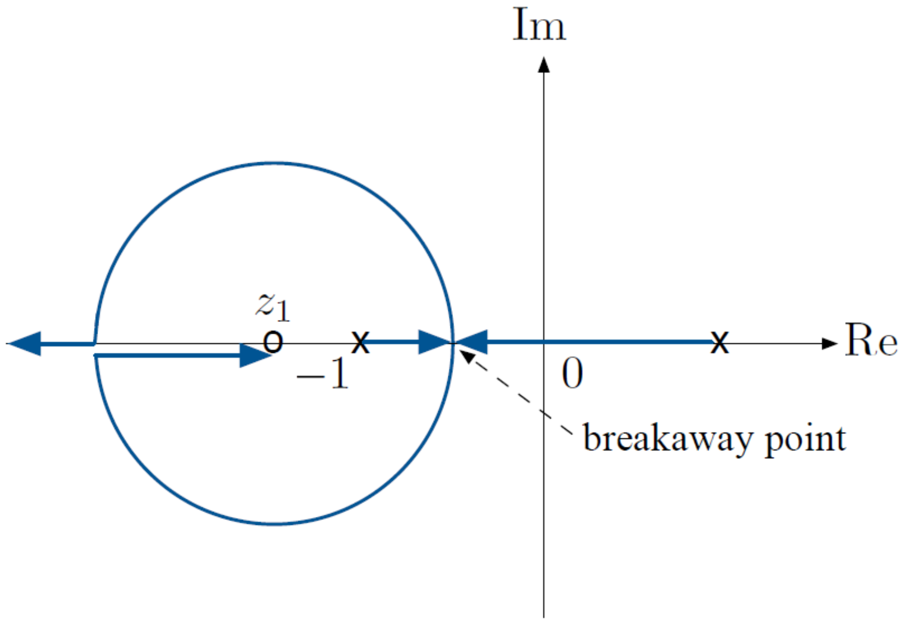


not be tested!

in textbook, not be tested

*__Theorem__: For the loop transfer function with one zero and two poles, then the root locus in the complex plane must be an arc with the zero as its center.

$$L(s) = \frac{s - z_1}{s^2 - 1}$$



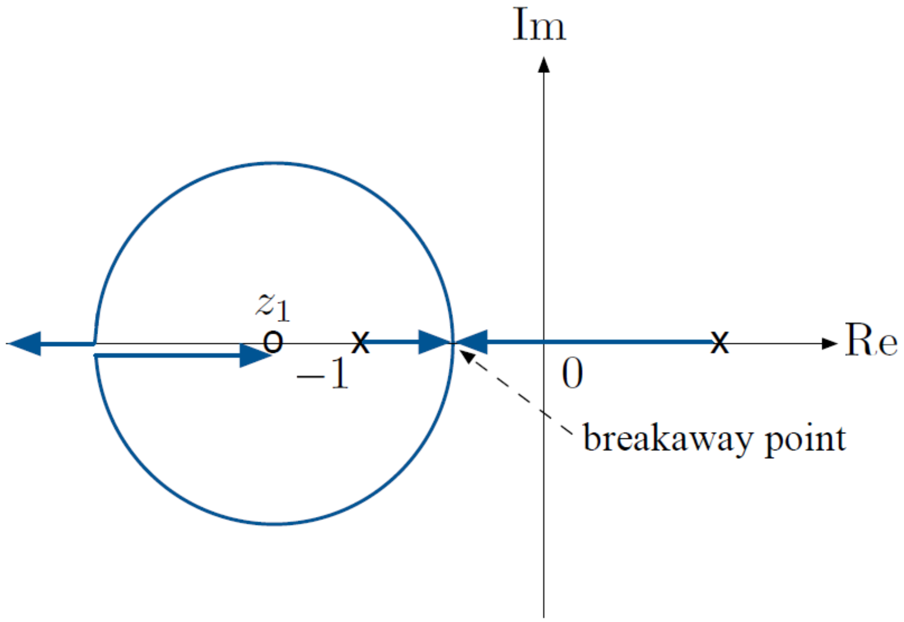
Why does one of the branches go off to $-\infty$?

$$s^2 - 1 + K(s - z_1) = 0$$

$$s^2 + Ks - (Kz_1 + 1) = 0$$

$$s = -\frac{K}{2} \pm \sqrt{\frac{K^2}{4} + Kz_1 + 1}, \;\; z_1 < 0 \qquad \text{as } K \to \infty, \; s \text{ will be} < 0$$

$$L(s) = \frac{s - z_1}{s^2 - 1}$$



Is the point $s = 0$ on the root locus?

Let's see if there is any value $K > 0$, for which this is possible:

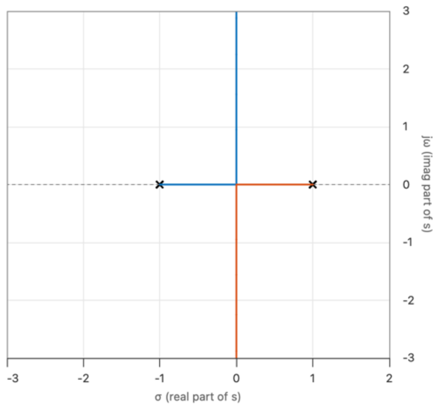
$$1 + KL(0) = 0$$

$$1 + Kz_1 = 0 \qquad K = -\frac{1}{z_1} > 0 \text{ does the job}$$

Main Points

- When zeros are in LHP, *high gain* can be used to stabilize the system (although one must worry about zeros at infinity).
- If there are zeros in RHP, high gain is always disastrous.
- PD control is effective for stabilization because it introduces a zero in LHP.

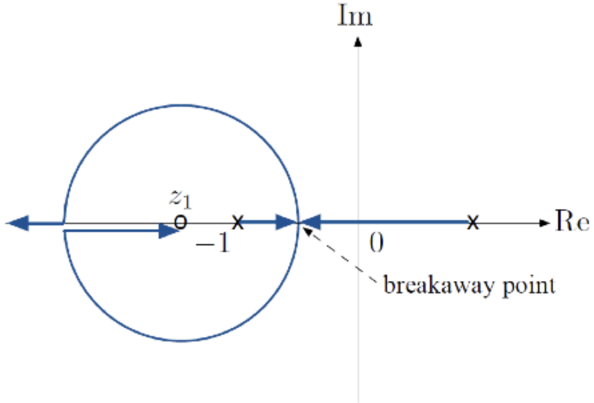
$$L(s) = \frac{1}{s^2 - 1}$$



$$\boxed{K_{\mathrm{P}} + K_{\mathrm{D}} s}$$

Add a zero on the LHP

$$L(s) = \frac{s - z_1}{s^2 - 1}$$



When **zeros are in the left-half plane**, they help "pull" the root locus into the **stable region**. This allows you to use **higher gains** without worrying about instability.

Main Points

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But: Rules A–C cannot tell the whole story. How do we know which way the branches go, and which pole corresponds to which zero?

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Rules D–F!!

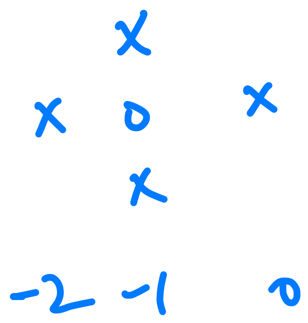
Example

Let's consider $L(s) = \dfrac{s+1}{s(s+2)((s+1)^2+1)}$

$z = -1$

$p = 0 \quad p = -2 \quad p = -1 \pm j$

branches = 4



Example

Let's consider $\boldsymbol{L(s) = \frac{s+1}{s(s+2)((s+1)^2+1)}}$

- Rule A: $\begin{cases} m = 1 \\ n = 4 \end{cases} \implies$ 4 branches

Example

Let's consider $L(s) = \dfrac{s+1}{s(s+2)((s+1)^2+1)}$

- Rule A: $\begin{cases} m = 1 \\ n = 4 \end{cases} \implies$ 4 branches
- Rule B: branches start at open-loop poles
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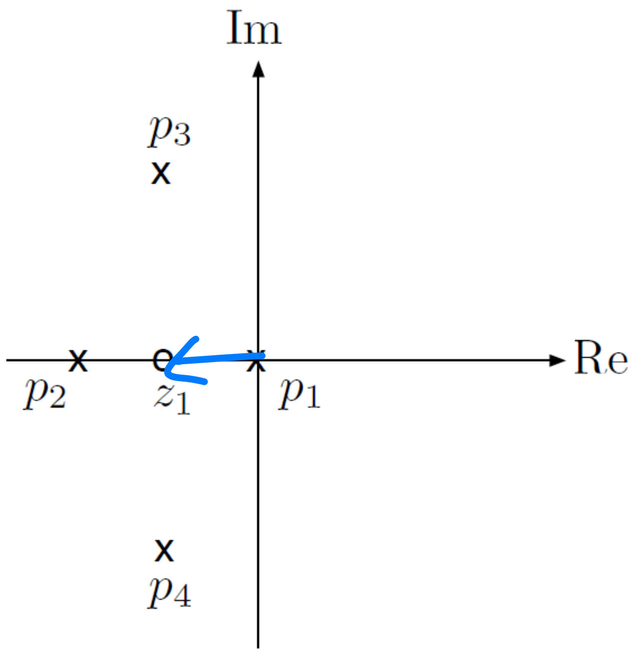
- Rule A: $\begin{cases} m = 1 \\ n = 4 \end{cases} \implies$ 4 branches

Starting points

- Rule B: branches start at open-loop poles
  $s = 0,\ s = -2,\ s = -1 \pm j$

ending points

- Rule C: branches end at open-loop zeros $\quad s = -1, \pm\infty$



Example, continued

Three more rules:

- Rule D: real locus
- Rule E: asymptotes
- Rule F: $j\omega$-crossings

Phase condition

$1 + KL = 0$

$L = -\frac{1}{K}$ (-ve real number)

$\angle L = 180^\circ$

Example, continued

Three more rules:

- Rule D: real locus
- Rule E: asymptotes
- Rule F: $j\omega$-crossings

Rules D and E are both based on the fact that

$$1 + KL(s) = 0 \text{ for some } K > 0 \quad \Longleftrightarrow \quad L(s) < 0$$

Rule D: Real Locus

The branches of the RL start at the open-loop poles. Which way do they go, left or right?

Recall the phase condition:

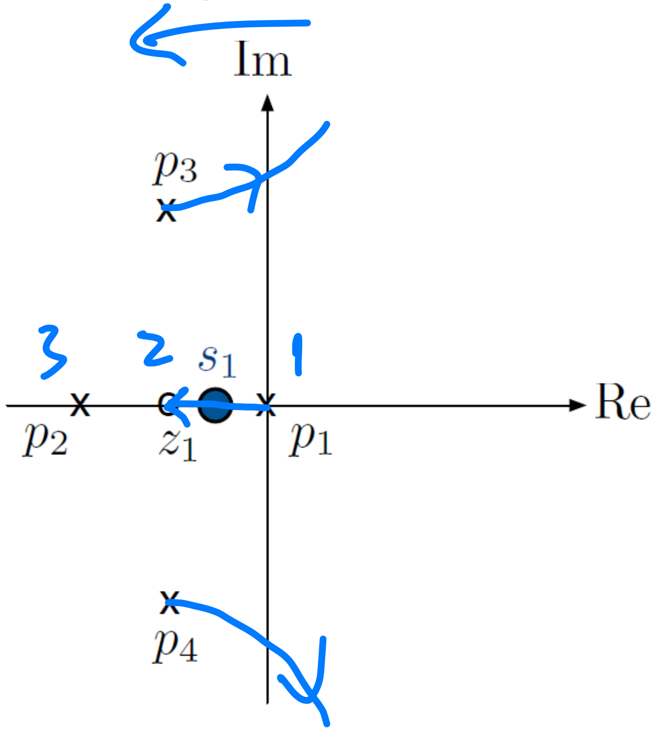
$$1 + KL(s) = 0 \qquad \Longleftrightarrow \qquad \angle L(s) = 180^\circ$$

$$\begin{aligned}
\angle L(s) &= \angle \frac{b(s)}{a(s)} \\
180^\circ \quad &= \angle \frac{(s-z_1)(s-z_2)\dots(s-z_m)}{(s-p_1)(s-p_2)\dots(s-p_n)} \\
&= \sum_{i=1}^{m} \angle(s-z_i) - \sum_{j=1}^{n} \angle(s-p_j)
\end{aligned}$$

— this sum must be $\pm 180^\circ$ for *any* s that lies on the RL.

Rule D: Real Locus

So, we try test points:



Rule D: Real Locus

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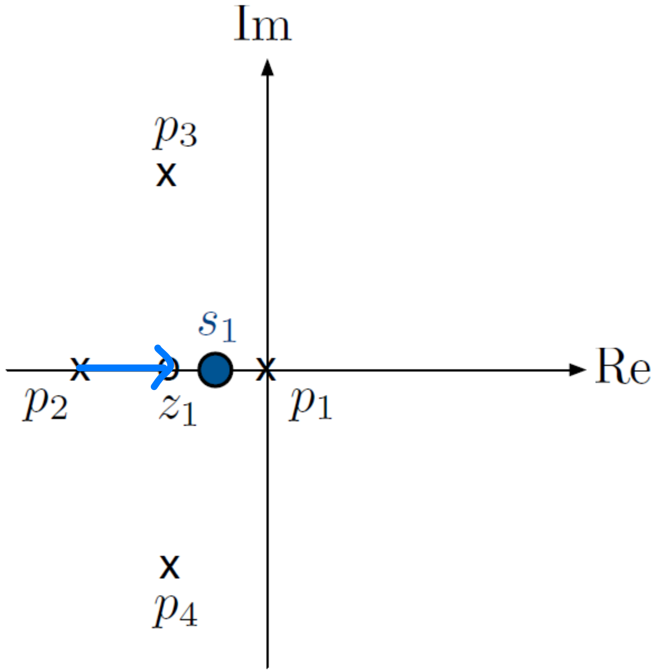


from ? to s_1

Example: p_2 to z_1

$$\angle(s_1 - z_1) = 0^\circ \quad (s_1 > z_1)$$
$$\angle(s_1 - p_1) = 180^\circ \quad (s_1 < p_1)$$
$$\angle(s_1 - p_2) = 0^\circ \quad (s_1 > p_2)$$
$$\angle(s_1 - p_3) = -\angle(s_1 - p_4)$$

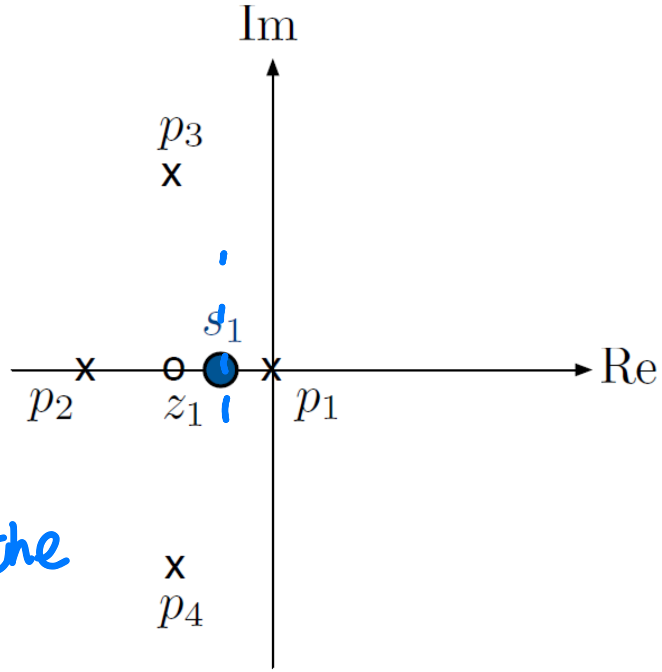
(conjugate poles cancel)

Rule D: Real Locus

So, we try test points:

$\angle L = \angle b - \angle a$

zeros   poles



Only need to care about poles/zeros on the real axis

$$\angle(s_1 - z_1) = 0^\circ \quad (s_1 > z_1)$$
$$\angle(s_1 - p_1) = 180^\circ \quad (s_1 < p_1)$$
$$\angle(s_1 - p_2) = 0^\circ \quad (s_1 > p_2)$$
$$\angle(s_1 - p_3) = -\angle(s_1 - p_4)$$

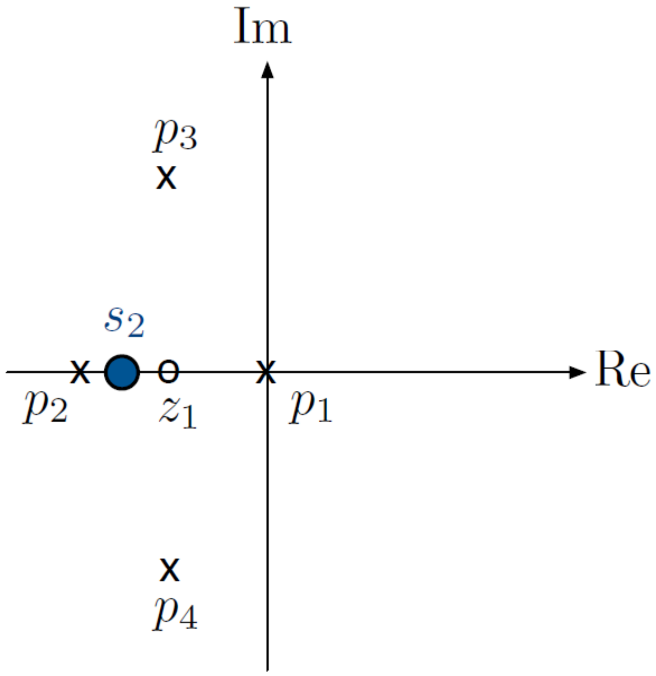
(conjugate poles cancel)

$$\angle(s_1 - z_1) - [\angle(s_1 - p_1) + \angle(s_1 - p_2) + \angle(s_1 - p_3) + \angle(s_1 - p_4)]$$
$$= 0^\circ - [180^\circ + 0^\circ + 0^\circ] = -180^\circ \qquad \checkmark\, s_1 \text{ is on RL}$$?

conflict → then what implement?

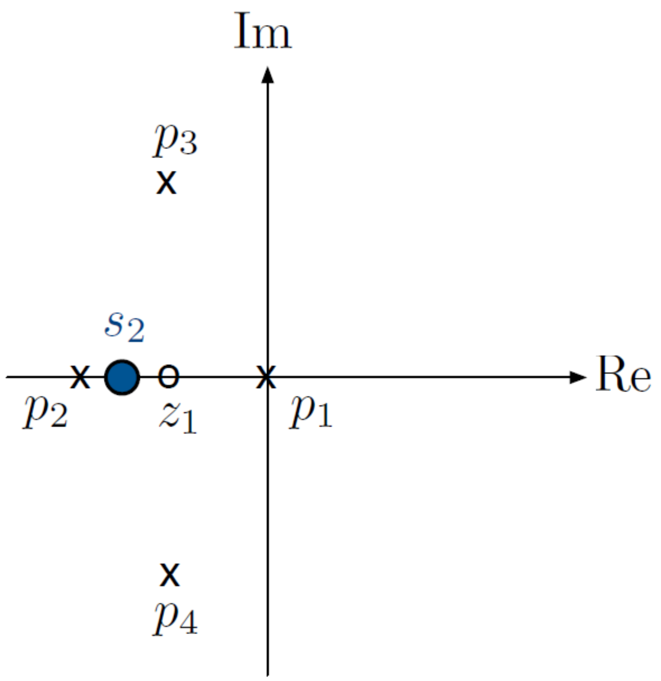
Rule D: Real Locus

Try more test points:



Rule D: Real Locus

Try more test points:



$$\angle(s_2 - z_1) = 180^\circ \quad (s_2 < z_2)$$
$$\angle(s_2 - p_1) = 180^\circ \quad (s_2 < p_1)$$
$$\angle(s_2 - p_2) = 0^\circ \quad (s_2 > p_2)$$
$$\angle(s_2 - p_3) = -\angle(s_1 - p_4)$$

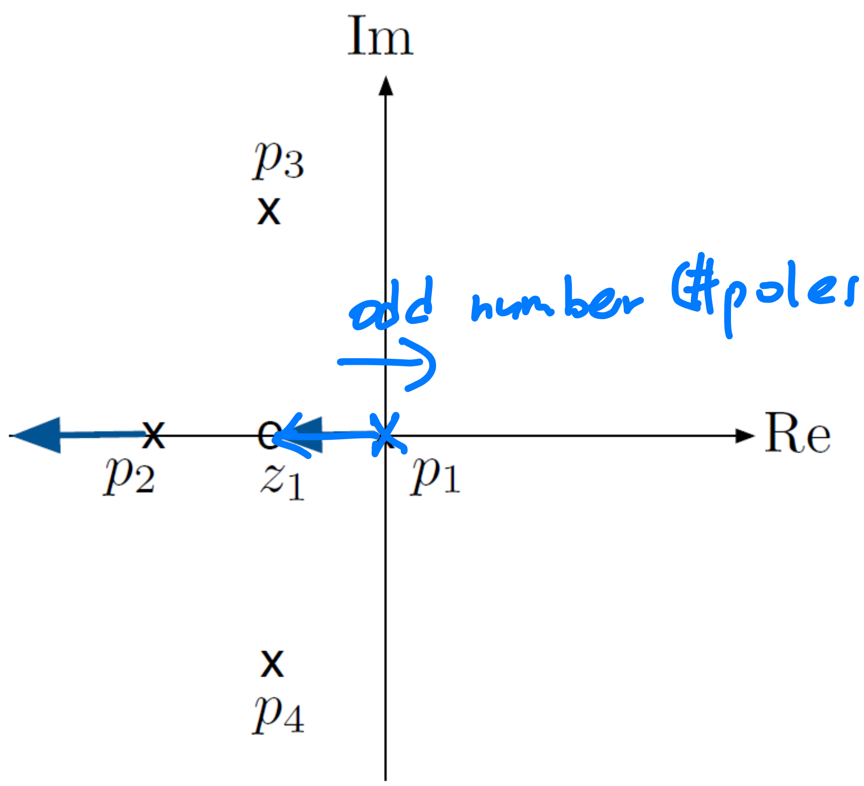
(conjugate poles cancel)

$$\angle(s_2 - z_1) - [\angle(s_2 - p_1) + \angle(s_2 - p_2) + \angle(s_2 - p_3) + \angle(s_2 - p_4)]$$
$$= 180^\circ - [180^\circ + 0^\circ + 0^\circ] = 0^\circ \qquad \times\ s_1 \text{ is not on RL}$$

Rule D: Real Locus

real line

Rule D: If s is *real*, then it is on the RL of $1 + KL$ if and only if there are an odd number of *real open-loop poles and zeros* to the right of s.



RL: Real Line

Rule E: Asymptotes

How does the locus look as $s \to \infty$?

final angle!

Rule E: Asymptotes

How does the locus look as $s \to \infty$?

$$180^\circ = \angle L(s) = \angle \frac{s^m + b_1 s^{m-1} + \dots}{s^n + a_1 s^{n-1} + \dots}$$

$$= \angle \frac{s^{m-n} + b_1 s^{m-n-1} + \dots}{1 + a_1 s^{-1} + \dots}$$

Approximation $\simeq \angle s^{m-n}$ if $|s| \to \infty$ (recall $m \leq n$)

Rule E: Asymptotes

How does the locus look as $s \to \infty$?

$$
\begin{aligned}
180^\circ = \angle L(s) &= \angle \frac{s^m + b_1 s^{m-1} + \ldots}{s^n + a_1 s^{n-1} + \ldots} \\
&= \angle \frac{s^{m-n} + b_1 s^{m-n-1} + \ldots}{1 + a_1 s^{-1} + \ldots} \\
&\simeq \angle s^{m-n} \;\text{ if } |s| \to \infty \qquad (\text{recall } m \leq n)
\end{aligned}
$$

Claim: If $\angle s^{m-n} = 180^\circ$, then

$$
\angle s = \frac{180^\circ + \ell \cdot 360^\circ}{n - m}, \qquad \ell = 0, 1, \ldots, n - m - 1
$$

Proof:

$n =$ $m =$

Rule E: Asymptotes

How does the locus look as $s \to \infty$?

$$\begin{aligned}180^\circ = \angle L(s) &= \angle \frac{s^m + b_1 s^{m-1} + \dots}{s^n + a_1 s^{n-1} + \dots} \\ &= \angle \frac{s^{m-n} + b_1 s^{m-n-1} + \dots}{1 + a_1 s^{-1} + \dots} \\ &\simeq \angle s^{m-n} \text{ if } |s| \to \infty \qquad (\text{recall } m \le n)\end{aligned}$$

Claim: If $\angle s^{m-n} = 180^\circ$, then

$$\angle s = \frac{180^\circ + \ell \cdot 360^\circ}{n-m}, \qquad \ell = 0, 1, \dots, n-m-1$$

Proof:

$$s = |s| e^{j\angle s} \qquad s^{m-n} = |s|^{m-n} e^{j(m-n)\angle s}$$

$$(m-n)\angle s = 180^\circ \qquad \Longrightarrow \qquad (m-n)\angle s = 180^\circ + \ell \cdot 360^\circ$$

Rule E: Asymptotes

Rule E: Branches near ∞ have phase

$$\begin{aligned}\angle s &\simeq \frac{180^\circ + \ell \cdot 360^\circ}{n-m} \\ &= \frac{(2\ell+1)\cdot 180^\circ}{n-m}, \qquad \ell = 0, 1, \ldots, n-m-1\end{aligned}$$

Note: if $m = n$, then there are no branches at ∞.

Back to Example: Rule E

Branches near ∞ have phase

$$\angle s = \frac{(2\ell + 1) \cdot 180^\circ}{n - m}, \qquad \ell = 0, 1, \ldots, n - m - 1$$

In our example, $\boldsymbol{L(s) = \dfrac{s+1}{s(s+2)((s+1)^2+1)}}$ $\quad \begin{cases} n = 4 \\ m = 1 \end{cases}$

$$\angle s = \frac{(2\ell + 1) \cdot 180^\circ}{3}, \qquad \ell = 0, 1, 2$$

$$\ell = 0: \qquad \frac{2 \cdot 0 + 1}{3} 180^\circ = 60^\circ$$

$$\ell = 1: \qquad \frac{2 \cdot 1 + 1}{3} 180^\circ = 180^\circ$$

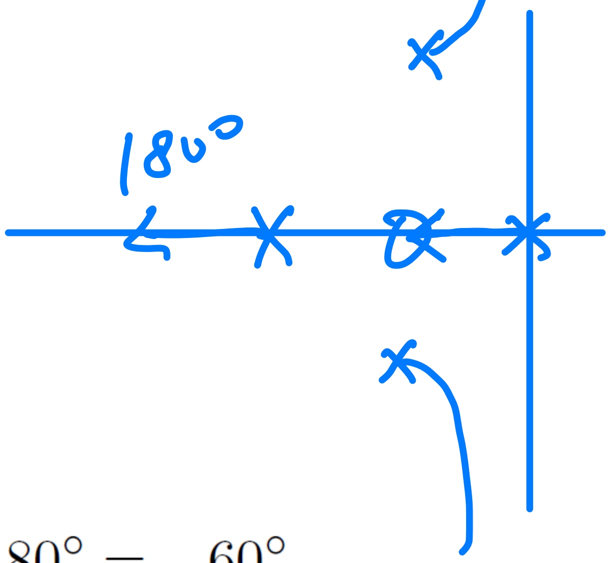
$$\ell = 2: \qquad \frac{2 \cdot 2 + 1}{3} 180^\circ = \frac{5}{3} 180^\circ = \left(2 - \frac{1}{3}\right) 180^\circ = -60^\circ$$

Back to Example: Rule E

Branches near ∞ have phase

$$\angle s = \frac{(2\ell + 1) \cdot 180^\circ}{n - m}, \qquad \ell = 0, 1, \ldots, n - m - 1$$

In our example, $\boldsymbol{L(s) = \dfrac{s+1}{s(s+2)((s+1)^2+1)}}$ $\begin{cases} n = 4 & \text{\# of poles} \\ m = 1 & \text{\# of zeros} \end{cases}$



$$\angle s = \frac{(2\ell + 1) \cdot 180^\circ}{3}, \qquad \ell = 0, 1, 2$$

$$\ell = 0: \qquad \frac{2 \cdot 0 + 1}{3} 180^\circ = 60^\circ$$

$$\ell = 1: \qquad \frac{2 \cdot 1 + 1}{3} 180^\circ = 180^\circ$$

$$\ell = 2: \qquad \frac{2 \cdot 2 + 1}{3} 180^\circ = \frac{5}{3} 180^\circ = \left(2 - \frac{1}{3}\right) 180^\circ = -60^\circ$$

会否 cross jω-axis?

Rule F

— asymptotes have angles 60°, 180°, -60°

Rule F: $j\omega$-crossings

Do the branches of the root locus cross the $j\omega$ axis?
(transition from *stability* to *instability*)

Rule F: $j\omega$-crossings

Do the branches of the root locus cross the $j\omega$ axis?
(transition from *stability* to *instability*)

Goal: determine if the equation

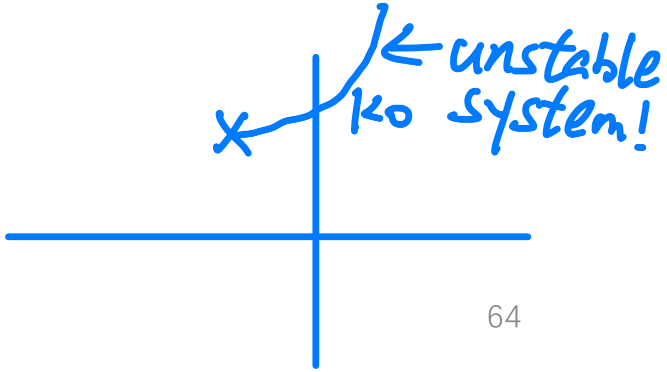
$$a(j\omega) + Kb(j\omega) = 0$$

has a solution $\omega \geq 0$ for some $K > 0$.

$$1 + \frac{kb}{a} = 0$$

$$a + kb = 0$$

$k > k_0 \Rightarrow$ unstable $\Rightarrow$



Rule F: $j\omega$-crossings

Do the branches of the root locus cross the $j\omega$ axis?
(transition from *stability* to *instability*)

Goal: determine if the equation

$$a(j\omega) + Kb(j\omega) = 0$$

has a solution $\omega \geq 0$ for some $K > 0$.

Best approach here: use the *Routh test* to first determine the critical value of K (when the characteristic polynomial becomes unstable), then plug it in and solve for $j\omega$-crossings (numerically or analytically).

Rule F: $j\omega$-crossings

In our example, the characteristic polynomial is

$$s^4 + 4s^3 + 6s^2 + (4+K)s + K$$

$$L(s) = \frac{s+1}{s(s^3+4s^2+6s+4)}$$

$$s^4 + 4s^3 + 6s^2 + 4s + K(s+1)$$

1	6	K
4	$4+K$	0
$\frac{20-K}{4}$	K	0
$\frac{(4+K)(20-K)-16K}{20-K}$	0	0
K	0	0

$$\frac{20-K}{4} > 0$$

$$20-K > 0$$

$$K < 20$$

$$(4+K)(20-K)-16K > 0$$

$$80 - 4K + 20K - K^2 - 16K > 0$$

$$80 - K^2 > 0$$

$$K^2 - 80 < 0$$

$$-\sqrt{80} < K < \sqrt{80}$$

Denominator can be scaled up, as we want the range to be positive!

Rule F: $j\omega$-crossings

In our example, the characteristic polynomial is

$$s^4 + 4s^3 + 6s^2 + (4 + K)s + K$$

K must be +ve

$K \in [0, \sqrt{80})$

$K_{critical} = \sqrt{80}$

Form the Routh array:

Also okay to scale it by multiplying +ve constants!

$$\begin{array}{lccc} s^4: & 1 & 6 & K \\ s^3: & 4 & 4+K & 0 \\ s^2: & 20-K & 4K & \\ s^1: & 80-K^2 & 0 & \\ s^0: & 4K & & \end{array}$$

For stability, need $20 - K > 0$, $80 - K^2 > 0$, $4K > 0$

The characteristic polynomial is stable for $K < \sqrt{80} = 4\sqrt{5}$

$$\implies K_{\text{critical}} = 4\sqrt{5}$$

Rule F: $j\omega$-crossings

$(j\omega)^4 + 4(j\omega)^3 + 6(j\omega)^2 + (4+\sqrt{80})(j\omega) + \sqrt{80} = 0$

In our example, the characteristic polynomial is

$$s^4 + 4s^3 + 6s^2 + (4+K)s + K$$

$\omega^4 - 4j\omega^3 - 6\omega^2 + (4+\sqrt{80})j\omega + \sqrt{80} = 0$

The critical value: $K = 4\sqrt{5}$ (from Routh test).

$\omega^4 - 6\omega^2 + \sqrt{80} = 0$

To find the $j\omega$-crossing, plug in and solve:

$-4\omega^2 - (4+\sqrt{80}) = 0$

$$(j\omega)^4 + 4(j\omega)^3 + 6(j\omega)^2 + (4+4\sqrt{5})j\omega + 4\sqrt{5} = 0$$

$$\omega^4 - 4j\omega^3 - 6\omega^2 + (4+4\sqrt{5})j\omega + 4\sqrt{5} = 0$$

real part: $\omega^4 - 6\omega^2 + 4\sqrt{5} = 0$ $\quad\quad$ $\omega^2 = 1+\sqrt{5}$

imag. part: $-4\omega^3 + 4(1+\sqrt{5})\omega = 0 \qquad \omega^2 = 1+\sqrt{5}$

$j\omega$-crossing at $j\omega_0 = \sqrt{1+\sqrt{5}} \approx 1.8$, when $K = 4\sqrt{5} \approx 8.9$

Why real part & imagine part are consistent?

Complete Root Locus

$$L(s) = \frac{s+1}{s(s+2)((s+1)^2+1)}$$

Rule A: 4 branches

Rule B: branches start at $p_1, \ldots, p_4$

Rule C: branches end at $z_1, \pm\infty$

Rule D: real locus $= [z_1, p_1] \cup (-\infty, p_2]$

Complete Root Locus

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$$\omega_0 = \sqrt{1+\sqrt{5}} \approx 1.8$$

when $K = 4\sqrt{5} \approx 8.9$

(transition from stability to instability)

Complete Root Locus

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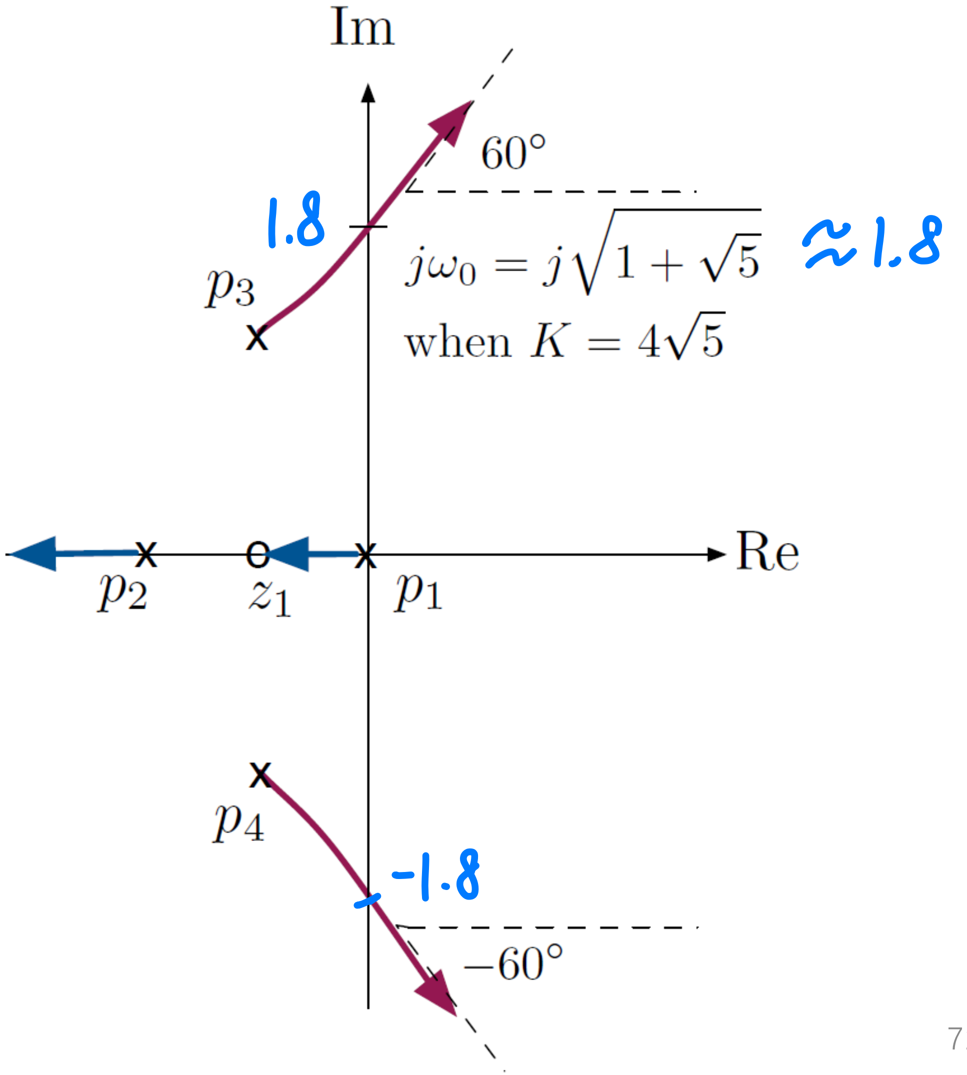
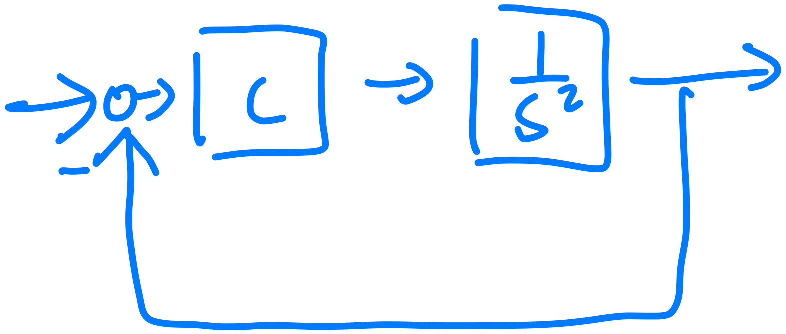


Control Design Using Root Locus

Case study: double integrator, transfer function $G(s) = \dfrac{1}{s^2}$

Control objective: ensure stability; meet time response specs.



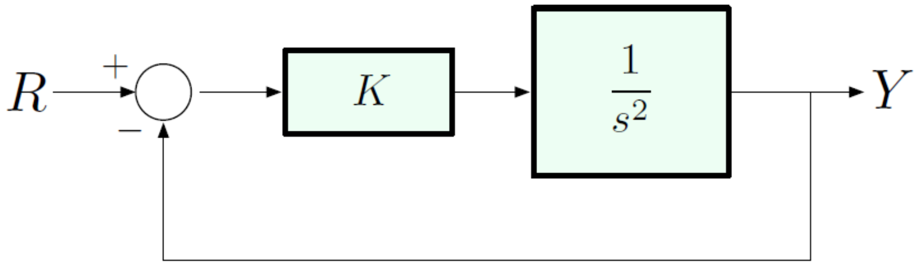
unstable system

Control Design Using Root Locus

Case study: double integrator, transfer function $G(s) = \dfrac{1}{s^2}$

Control objective: ensure stability; meet time response specs.

First, let's try a simple P-gain:



Closed-loop transfer function:

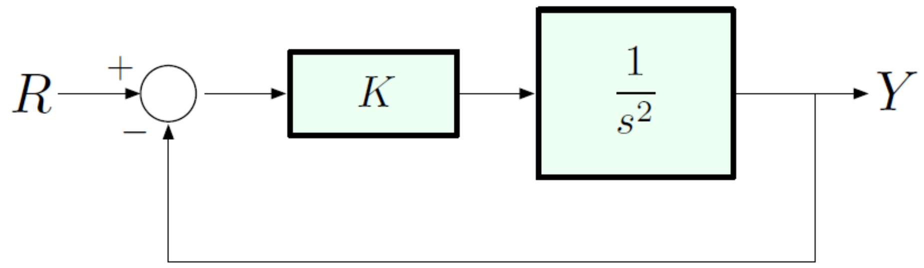
$$\frac{\frac{K}{s^2}}{1+\frac{K}{s^2}} = \frac{K}{s^2+K}$$

$$T(s) = \frac{\frac{K}{s^2}}{1+\frac{K}{s^2}} = \frac{K}{s^2+K}$$

$+0s \Rightarrow$ unstable

poles: $s = \pm\sqrt{K}\,j$

Double Integrator with P-Gain



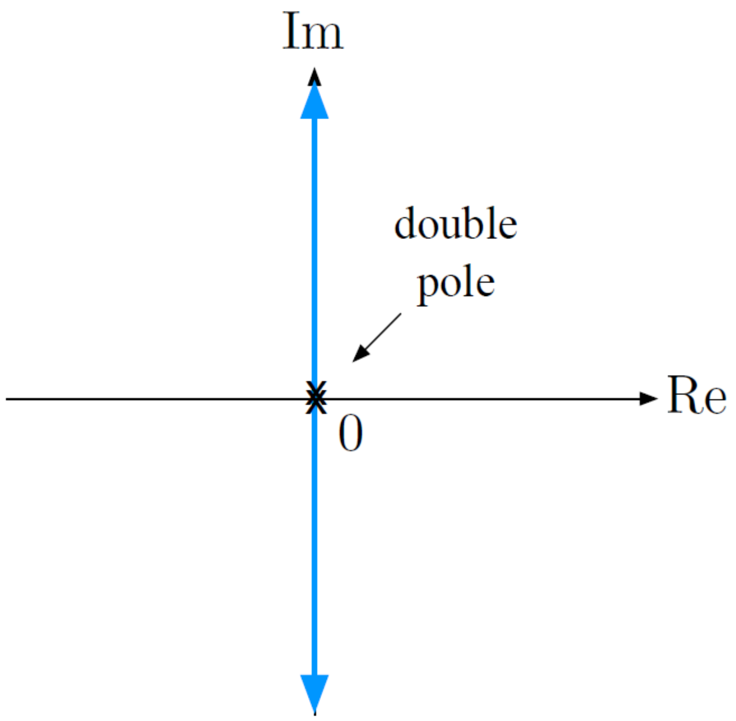
Closed-loop transfer function:

$$\frac{\frac{K}{s^2}}{1+\frac{K}{s^2}} = \frac{K}{s^2+K}$$

Characteristic equation:

$$s^2 + K = 0$$

Closed-loop poles: $s = \pm\sqrt{K}j$



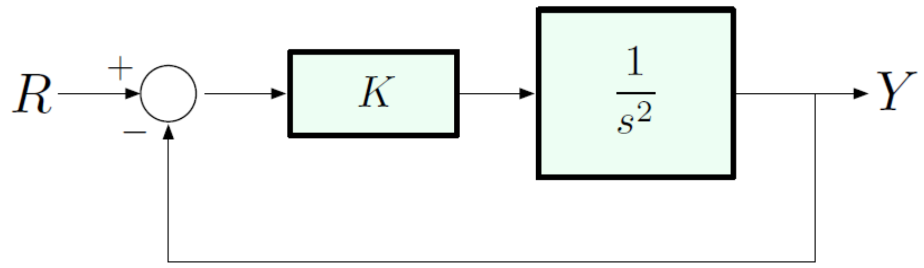
☆ 2 branches
2 starting point

0 ending points

Double Integrator with P-Gain

$s^2 + K = 0$

$\Rightarrow$ no $j\omega$ - crossing



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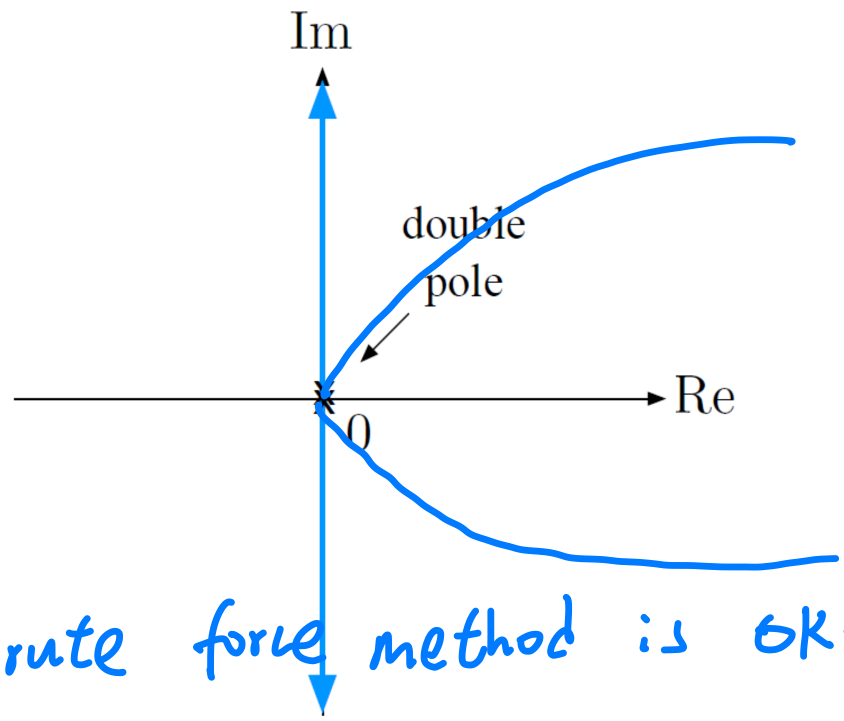
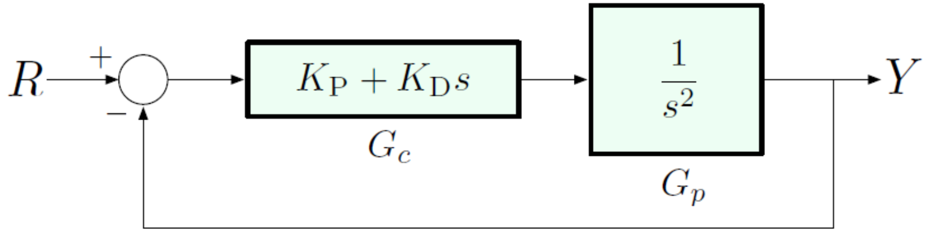


(Brute force method is OK!)

This confirms what we already knew: P-gain alone does not deliver stability.

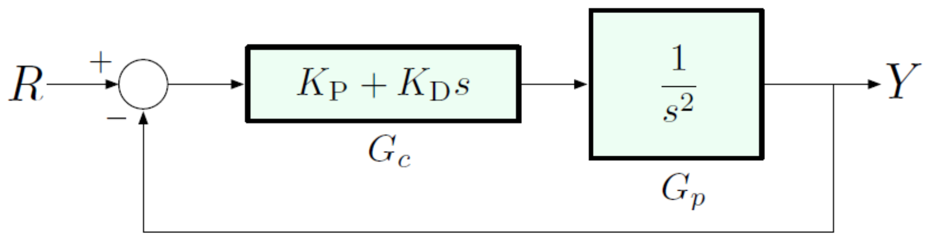
Double Integrator with PD-Control



Characteristic equation: $$1 + \underbrace{(K_\mathrm{P} + K_\mathrm{D}s)}_{G_c(s)} \cdot \underbrace{\frac{1}{s^2}}_{G_p(s)} = 0$$

$$s^2 + K_\mathrm{D}s + K_\mathrm{P} = 0$$

Double Integrator with PD-Control



$K_p + K_D s$

☆ Make highest order to be 1

$CP = KL$

$CP = \frac{K_p + K_D s}{s^2} = \frac{K_D\left(s + \frac{K_p}{K_D}\right)}{s^2}$

$\Rightarrow K = K_D$

Characteristic equation: $$1 + \underbrace{(K_P + K_D s)}_{G_c(s)} \cdot \underbrace{\frac{1}{s^2}}_{G_p(s)} = 0$$

$$s^2 + K_D s + K_P = 0$$

To use the RL method, we need to convert it into the Evans form $1 + KL(s) = 0$, where $L(s) = \frac{b(s)}{a(s)} = \frac{s^m + b_1 s^{m-1} + \ldots}{s^n + a_1 s^{n-1} + \ldots}$

$$1 + (K_P + K_D s)\frac{1}{s^2} = 1 + K_D \cdot \frac{s + K_P/K_D}{s^2}$$

$$\Longrightarrow K = K_D, \ L(s) = \frac{s + K_P/K_D}{s^2} \qquad \text{(assume } K_P/K_D \text{ fixed, } = 1)$$

Double Integrator with PD-Control

Characteristic equation: $\quad 1 + K \cdot \dfrac{s+1}{s^2} = 0$

Here we can still write out the roots explicitly:

$$s^2 + Ks + K = 0 \qquad \Longrightarrow \qquad s = \frac{-K \pm \sqrt{K^2 - 4K}}{2}$$

But let's actually draw the RL using the rules:

2 branches

2 poles

Double Integrator with PD-Control

Why Circle?

Appendix, will not in final exam!

Characteristic equation: $1 + K \cdot \frac{s+1}{s^2} = 0$

Here we can still write out the roots explicitly:

$$s^2 + Ks + K = 0 \qquad \Longrightarrow \qquad s = \frac{-K \pm \sqrt{K^2 - 4K}}{2}$$

But let's actually draw the RL using the rules:

Rule A: 2 branches

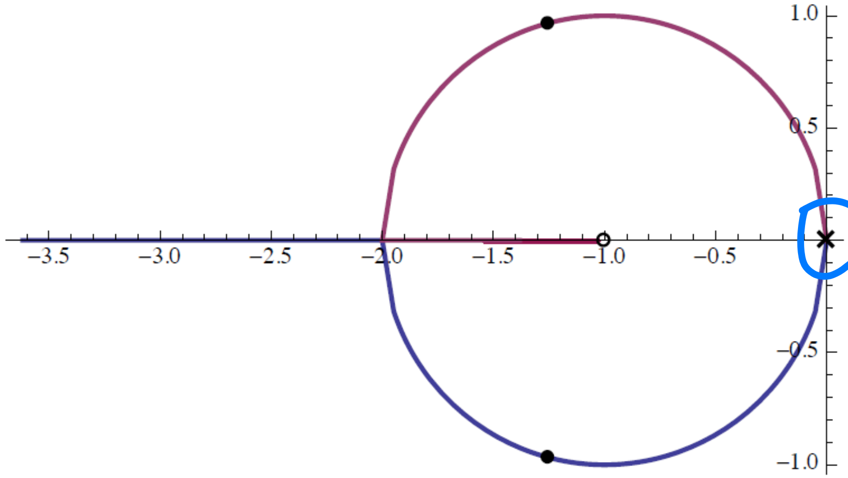
Rule B: both start at $s = 0$

Rule C: one ends at $z_1 = -1$, the other at ∞

Rule D: one branch will go off to $-\infty$

Rule E: asymptote angles at 180°

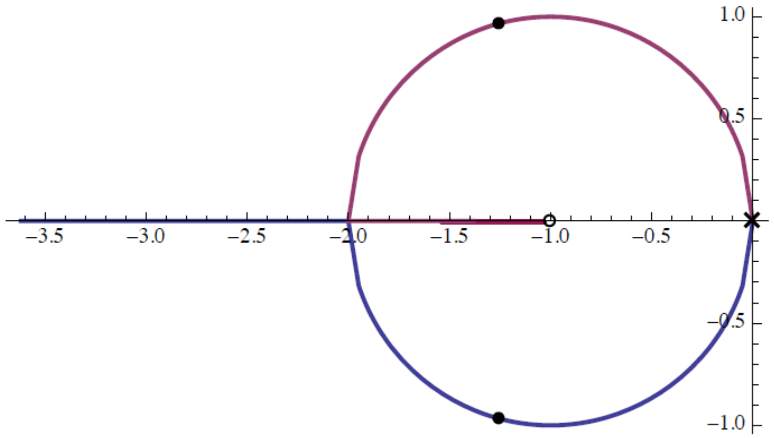
Rule F: no $j\omega$-crossings except for $s = p_1 = p_2 = 0$



Ignore this point! As KD > 0

Double Integrator with PD-Control

Characteristic equation: $1 + K \cdot \dfrac{s+1}{s^2} = 0$



The root locus stabilize the system!!!

What can we conclude from this root locus about stabilization?

- all closed-loop poles are in LHP (we already knew this from Routh, but now can visualize)
- nice damping, so can meet reasonable specs

So, the effect of D-gain was to introduce an *open-loop zero* into LHP, and this zero "pulled" the root locus into LHP, thus stabilizing the system.

Summary: General Guidelines In Root-locus Design

- Find the desired **dominant pole regions** from the transient requirements, e.g., t_r, M_p, t_p, t_s;
- Find the desired **steady-state error constants** from the steady-state tracking requirements;
- Decide **whether you need to increase the system type** in order to satisfy the design specifications: if the steady-state accuracy demands the increase of the system type, then PI or PID controllers must be used;
- Plot the **root locus of KP(s)**;
- **Check** whether the desired dominant poles can be selected on the root-locus plot: if yes, PI or phase-lag controller* can be used; if not, PD, PID, phase-lead*, or lead-lag* controllers must be used.

* This topic is not covered in this course. It you are interested in these controllers, you may find more details in Chapter 5 of our textbook. No need to care!

Another Example: Root Locus method for PID controller design

Question: Design a PID controller for plant:

$$G(s) = \frac{1}{s(s+2)}$$

$C(s) = K_P + K_D s + \frac{K_I}{s}$

to achieve the following objective:

- Rise time $t_r < 1$ second
- Overshoot $M_p < 10\%$
- Zero steady-state error (already satisfied due to integrator)

Another Example: Root Locus method for PID controller design

Solution:

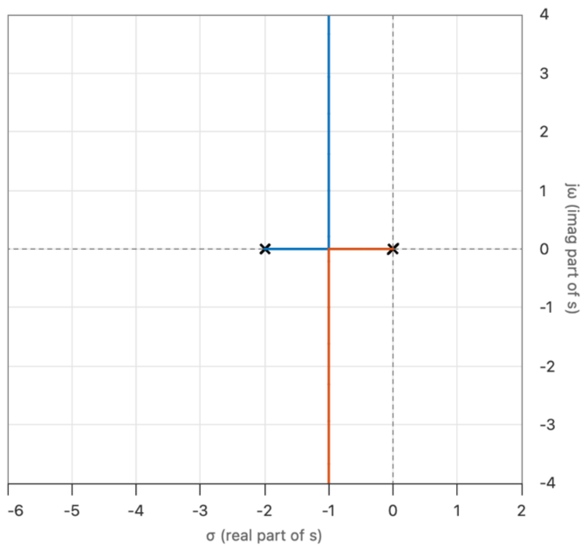
First, let's analysis this plant simply:

$$G(s) = \frac{1}{s(s+2)}$$

- This is a second-order, **Type 1** system (because of the pole at the origin). It already has **zero steady-state error to a step input.**
- By using Root Locus method, we can find:
  - Its response is likely slow (**integrator**, one pole is at s=0)
  - It could have significant overshoot (**low-damping** complex, as gain K increases, the closed-loop poles move along a curved path into the complex plane).

significant overshoot for large K!

  - For example, when k=4, overshoot=16%, settling time t_s=4s.



So, let's try to design a controller for it!

Another Example: Root Locus method for PID controller design

Solution (continue…):

$a_1 = 2\zeta\omega_n$

$a_0 = \omega_n^2$

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

We want to design a controller (like PID) so that the **closed-loop response** satisfies these goals (t_s, M_p).

- Rise time $t_r < 1$ second
- Overshoot $M_p < 10\%$

To proceed, we use **standard second-order system approximations**, because most systems are dominated by a pair of complex conjugate poles (called **dominant poles**).

- The percent overshoot M_p for a second-order system is related to damping ratio ζ by:

$$M_p = e^{\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)} \times 100\% \quad = 0.1$$

Set $M_p = 10\%$:

$$0.10 = e^{\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)} \Rightarrow \zeta \approx 0.591$$

So, we'll use $\zeta \approx 0.6$ as a good estimate.

- Approximate rise time for a second-order system:

$$t_r \approx \frac{1.8}{\omega_n}$$

Given $t_r < 1$ sec:

$$\frac{1.8}{\omega_n} < 1 \Rightarrow \omega_n > 1.8$$

$$\Rightarrow \quad s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -1.08 \pm j1.44$$

make the system to have this pole!

Another Example: Root Locus method for PID controller design

Solution (continue···):

Let's check the **Original Root Locus**:

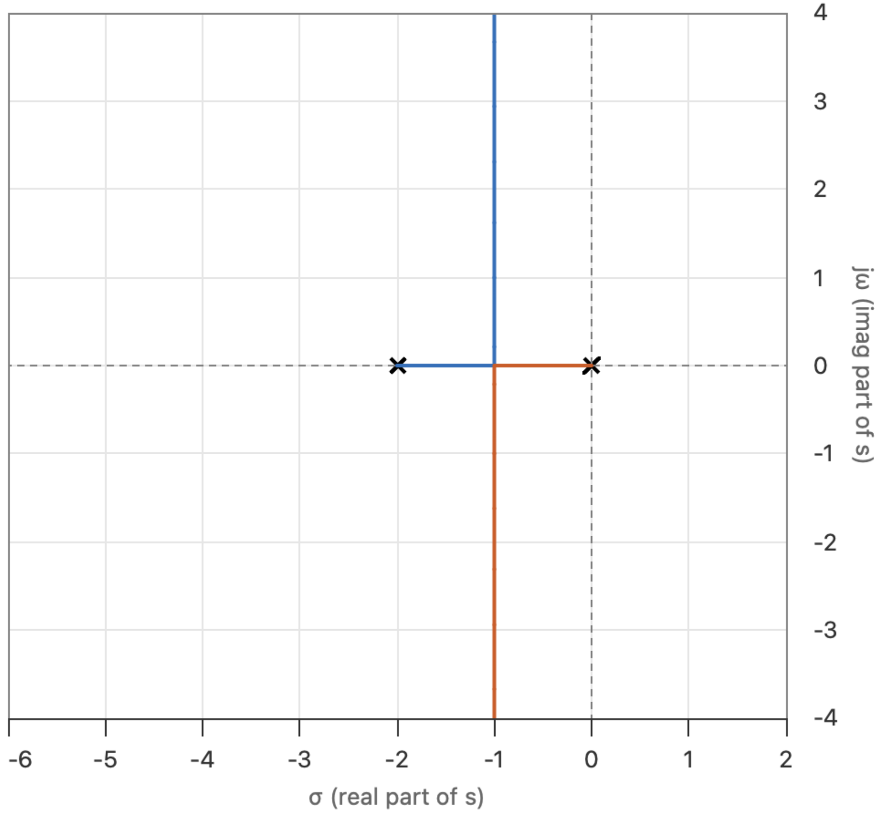
You'll notice: **The root locus doesn't pass through the desired poles:**

$$s = -1.08 \pm j1.44$$

So, adjusting gain K alone isn't enough.

Let's introduce a PID controller:

$$D(s) = K_p + \frac{K_i}{s} + K_d s = K_d \cdot \frac{s^2 + \frac{K_p}{K_d} s + \frac{K_i}{K_d}}{s}$$



Another Example: Root Locus method for PID controller design

Solution (continue…):

The PID controller:

$$D(s) = K_p + \frac{K_i}{s} + K_d s = K_d \cdot \frac{s^2 + \frac{K_p}{K_d} s + \frac{K_i}{K_d}}{s}$$

This means the PID controller adds:

- A **pole at the origin** (integrator – already exists in the plant)
- **Two zeros**, whose positions are determined by Kp, Ki and Kd

So, we can **shape the root locus** by placing these zeros smartly!

Another Example: Root Locus method for PID controller design

$\frac{1}{s(s+2)}$

Solution (continue···):

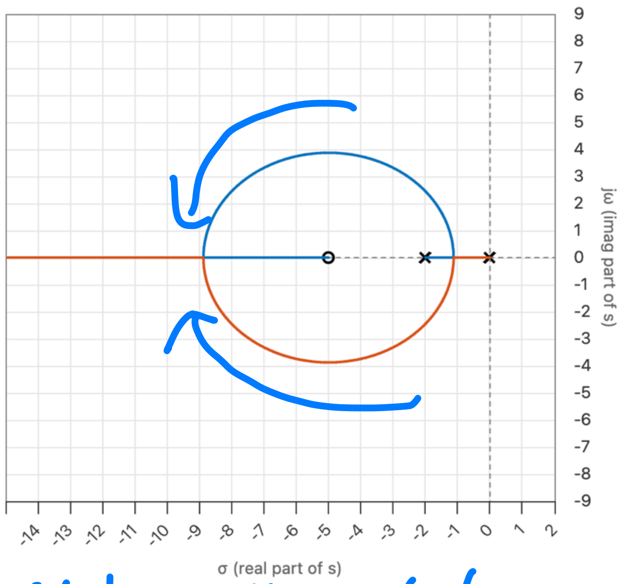
Since the system already has an integrator, we start with a **PD controller** (a simplified PID):

$$D(s) = K_p + \cancel{\frac{K_i}{s}} + K_d s$$

$$= K_p + K_d s$$

$$= K_d(s + K_p/K_d)$$ --- *Let's define: $K=K_d$, $z=K_p/K_d$.*

$$= K(s + z)$$ --- *This adds a* ***zero at*** *$s = -z$.*



Let's **choose z = 5** (you may try other values by yourself to see how the root-locus changes), so the open-loop system becomes:

$$G_{new}(s) = \frac{K(s+5)}{s(s+2)}$$

Note: the original plant is: $G(s) = \frac{1}{s(s+2)}$

Make the root locus close to the desired dominant poles!

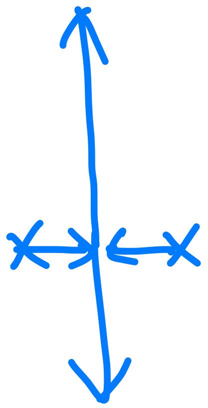
Now when we plot the root locus, it bends more toward the desired region (near $-1.08 \pm j1.44$). By **adjusting K**, we can **place the dominant poles at or near the desired location**.

Finally, we can get K_p and K_d by: $K_p = z * K_d$, $K_d = K$. *Done!*

Review & Exercise

- The concept of RL method?
- How to sketch the RL curve?
- How to use RL method to design the pole?
- Exercises:

Exercise 1



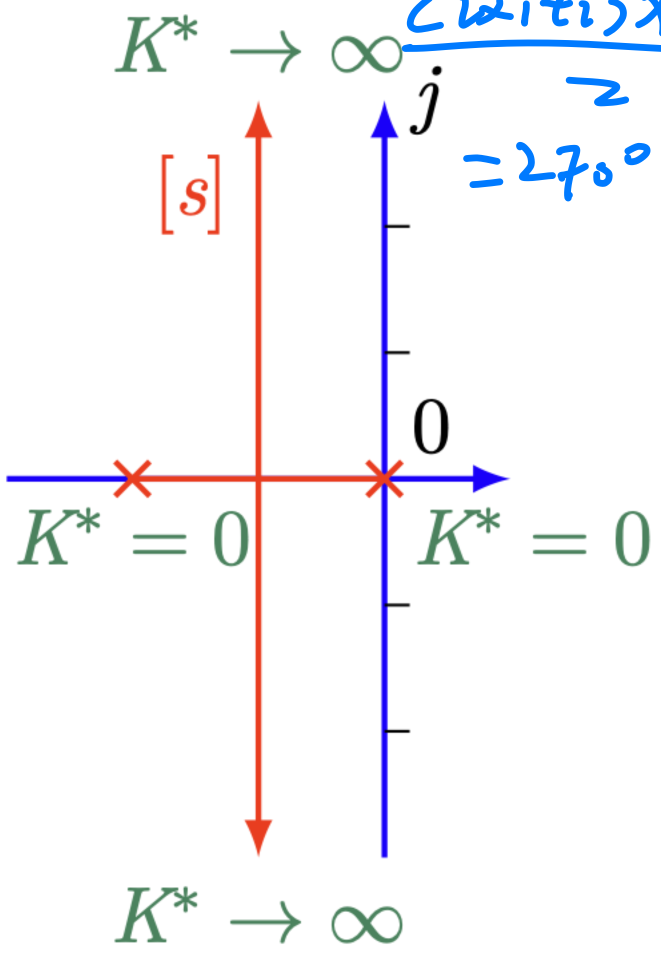
$m-n-1=1$

$m-n=2$

$$\frac{(2\times0+1)\times180^\circ}{2} = 90^\circ$$

$$\frac{(2\times1+1)\times180^\circ}{2} = 270^\circ$$

$$G(s)H(s) = \frac{K^*}{s(s+2)}$$

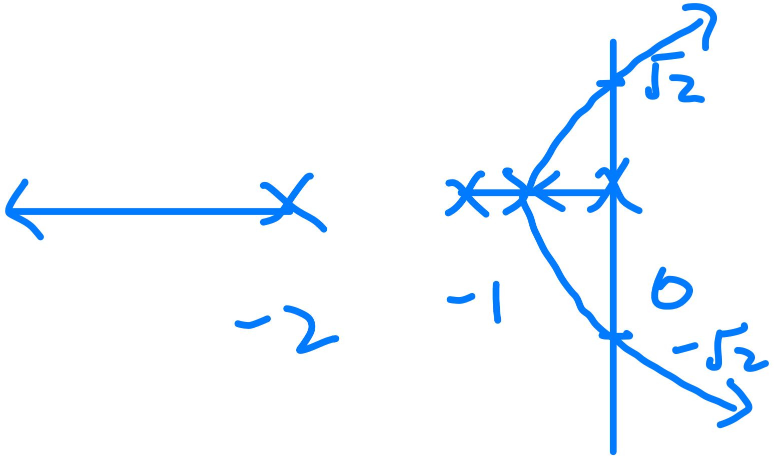


$s^2+2s+K=0$

$$s = \frac{-2 \pm \sqrt{4-4K}}{2}$$

$$= -1 \pm \sqrt{1-K}$$

Exercise 2



$3-0-1=2$ $\quad$ $60°,\ 180°,\ 300°$

$(0,1,2)$

$$G(s) = \frac{K^*}{s(s+1)(s+2)}$$

$-j\omega^3 - 3\omega^2 + 2j\omega + 6 = 0$ $\quad$ $-\omega^2 + 2 = 0$

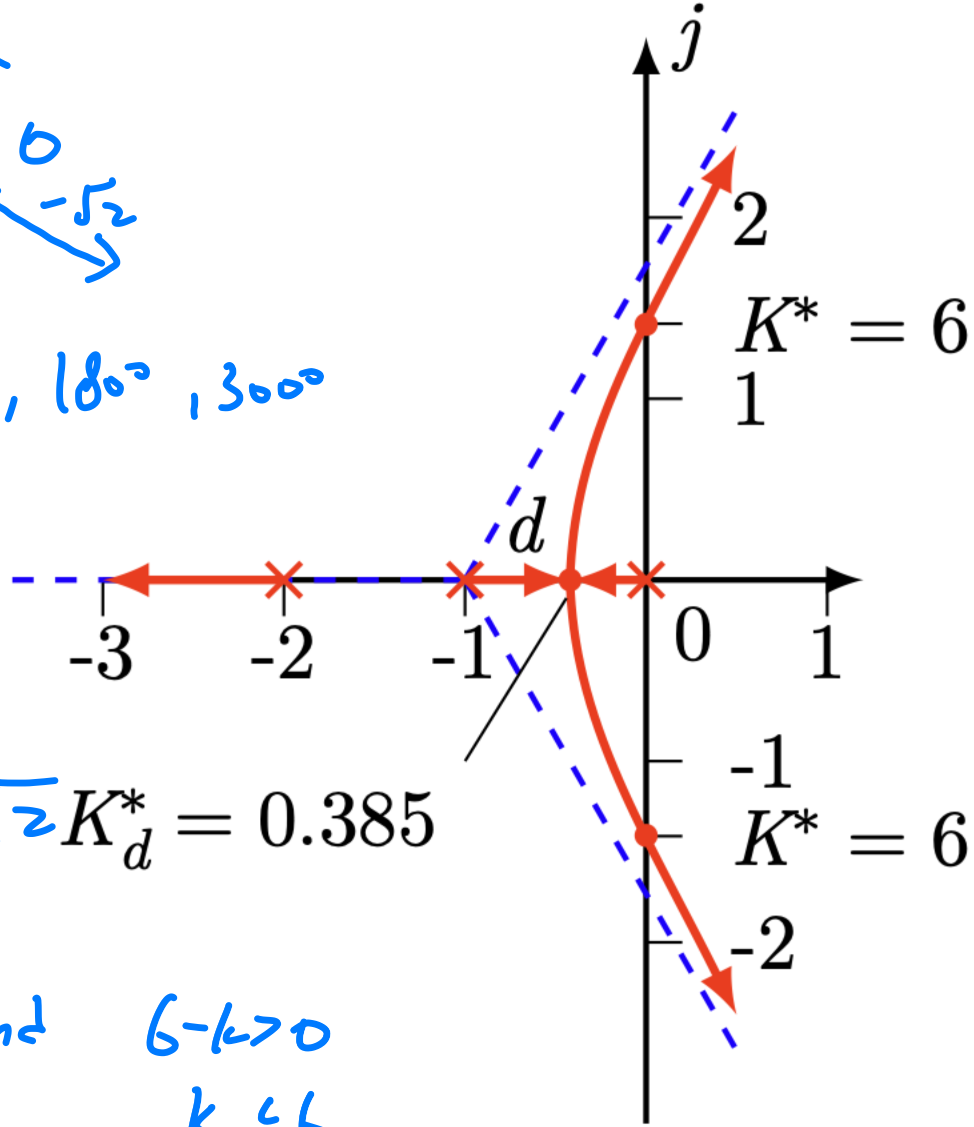
$\omega = \pm\sqrt{2}$

$s^3 + 3s^2 + 2s + K$

s^3	1	2
s^2	3	K
s	$\frac{6-K}{3}$	0
s^0	K	0

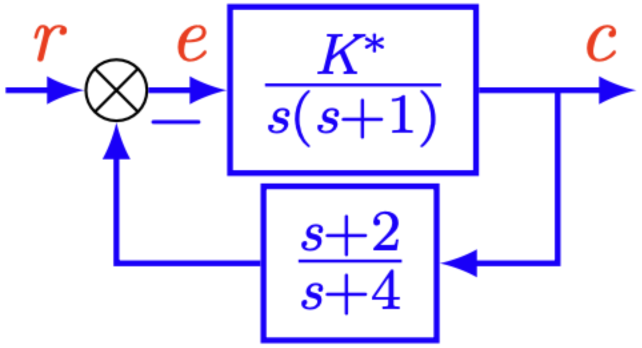
$K > 0$ and $6 - K > 0$

$K < 6$



Exercise 3

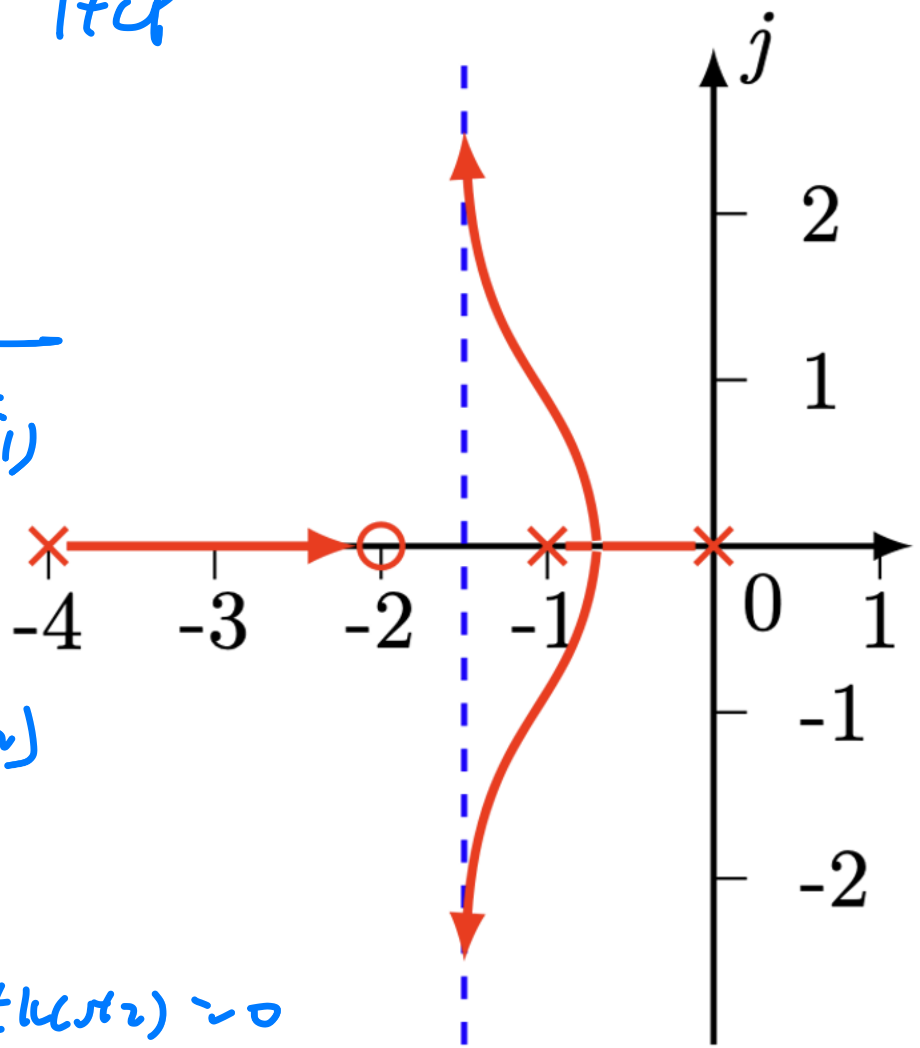
$$\frac{C}{1+CP} \qquad \frac{CP}{1+CP}$$



$$\frac{\frac{K}{s(s+1)}}{1+\frac{s+2}{s+4}\frac{K}{s(s+1)}}$$

$$= \frac{K(s+4)}{s(s+4)(s+1)+K(s+2)}$$

$$G(s) = \frac{K^*(s+2)}{s(s+1)(s+4)}$$



$$s(s+4)(s+1)+K(s+2)=0$$

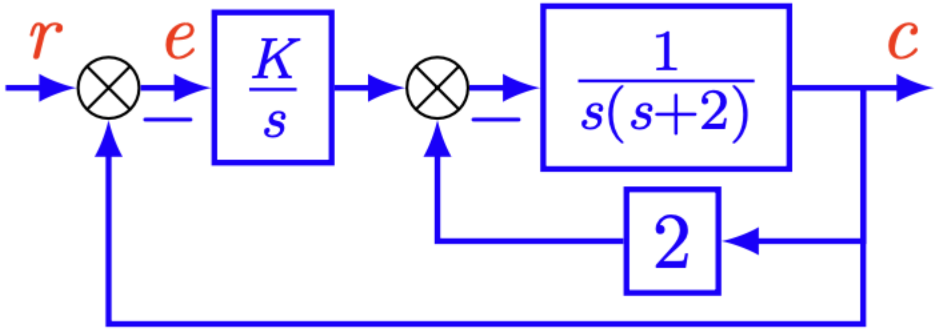
$$1+\frac{K(s+2)}{s(s+4)(s+1)}=0$$

Exercise 4

$$\frac{\frac{1}{s(s+2)}}{1+\frac{2}{s(s+2)}} = \frac{1}{s(s+2)+2}$$

$$G(s) = \frac{K}{s(s^2+2s+2)}$$

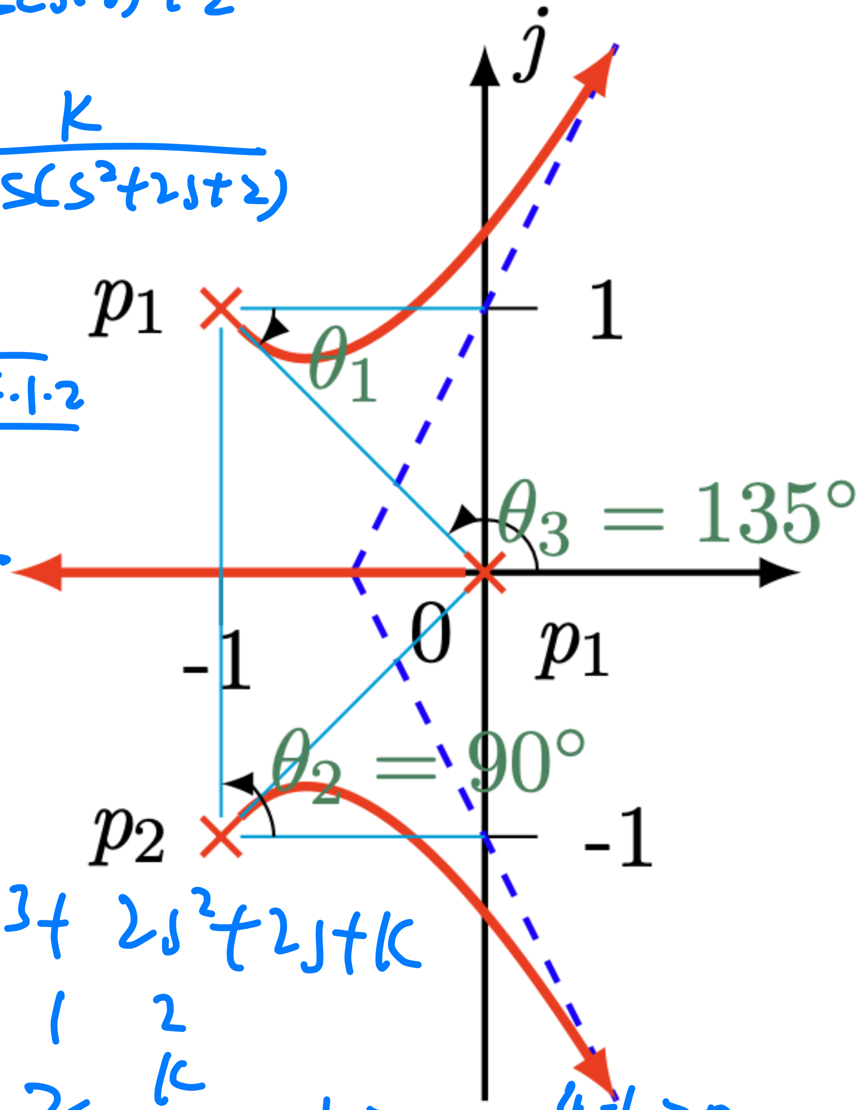
$$= \frac{-2\pm\sqrt{4-4\cdot1\cdot2}}{2} = \frac{-2\pm2j}{2} = -1\pm j$$

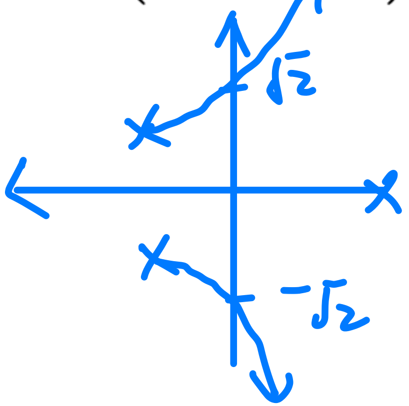


$$-j\omega^3 - 2\omega^2 + 2j\omega + 4$$

$$-\omega^2+2=0 \qquad \omega^2 = 2$$

$$G(s) = \frac{K}{s}\,\frac{\frac{1}{s(s+2)}}{1+\frac{2}{s(s+2)}} = \frac{K}{s(s^2+2s+2)}$$





$k \in \{0,1,2\}$

$\theta = 60^\circ,\ 180^\circ,\ 300^\circ$

s^3+2s^2+2s+K

1	2
2	K
$\frac{4-K}{2}$	0
K	0

$K>0$

$4-K>0$

$K<4$

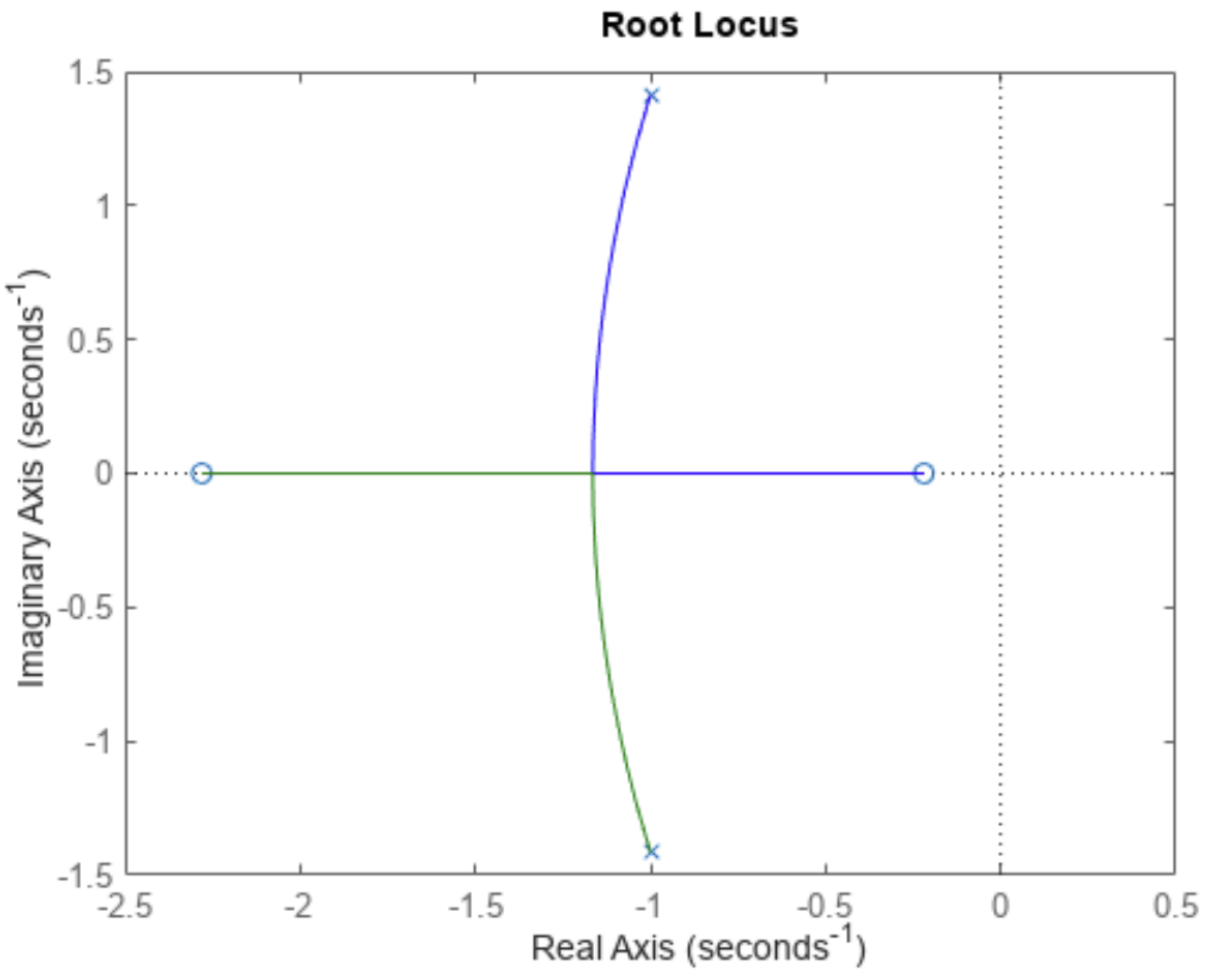
Appendix

Root locus by Matlab

$$sys(s) = \frac{2s^2 + 5s + 1}{s^2 + 2s + 3}.$$

Matlab Code:

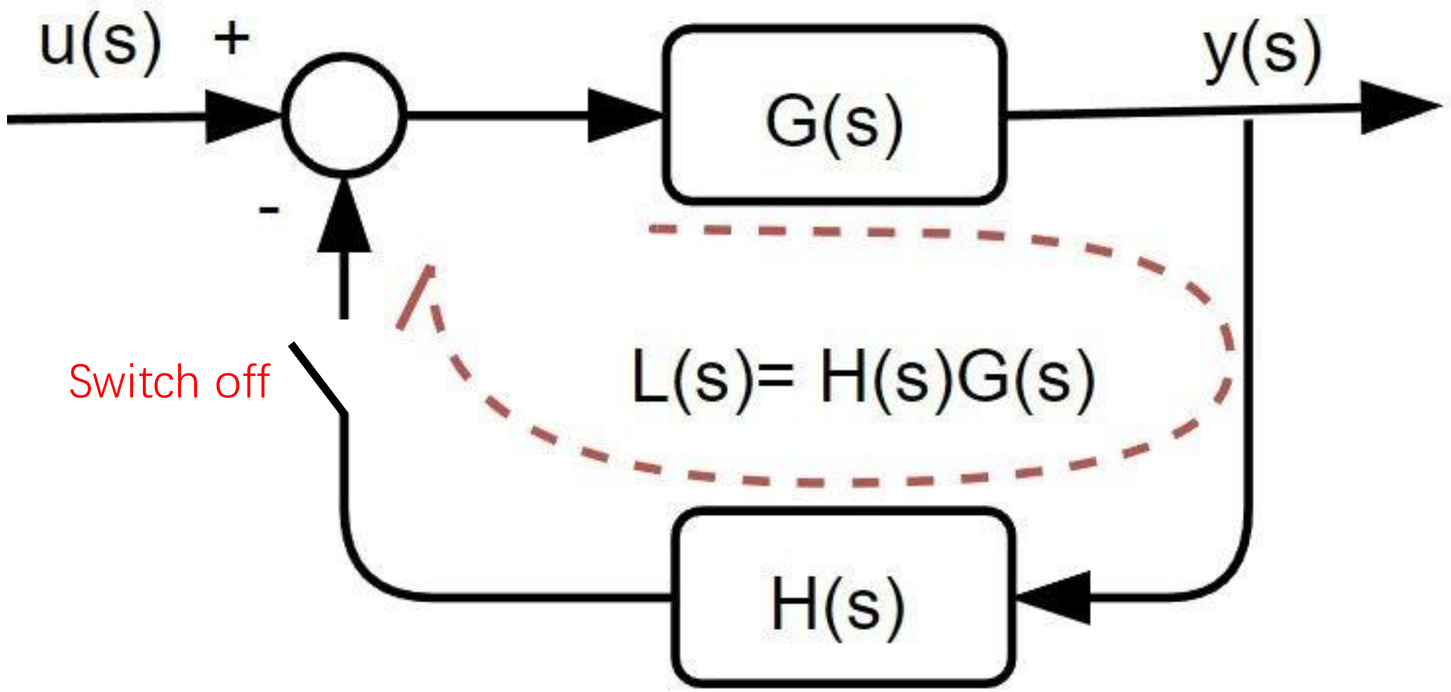
```
sys = tf([2 5 1], [1 2 3]);

rlocus(sys)
```



One website for Root Locus drawing

https://lpsa.swarthmore.edu/Root_Locus/RLDraw.html

Concept – Loop transfer function



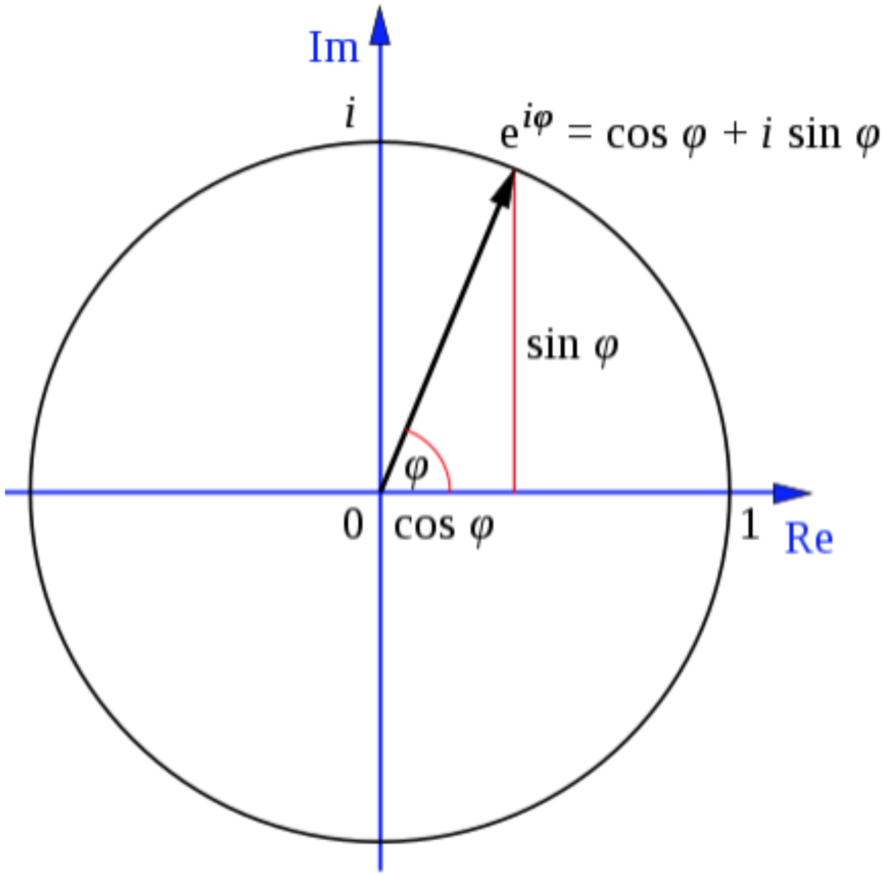
$$T(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s) \cdot H(s)} = \frac{G(s)}{1 + L(s)}$$

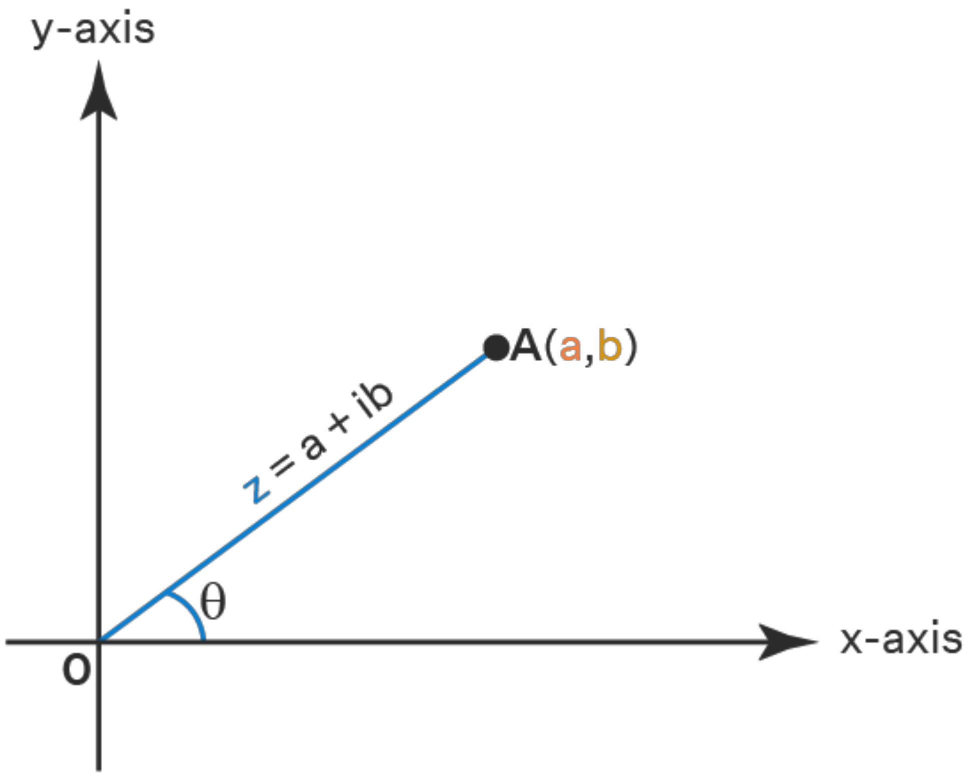
$$L(s) = \text{loop gain} = G(s) \cdot H(s)$$

T(s) : Closed-loop transfer function
L(s) : Loop transfer function

Note: In some textbook, the “loop transfer function” is also called as “open loop transfer function”.

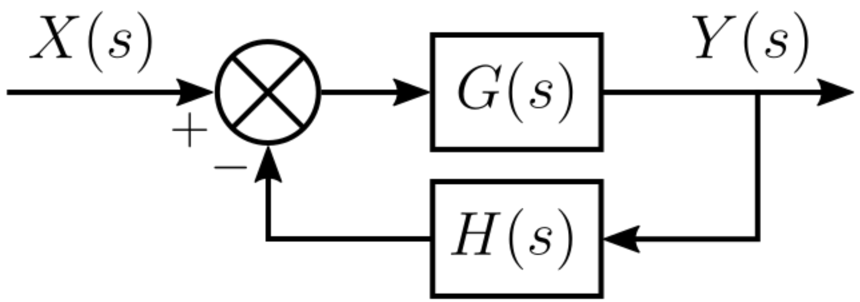
Concept: Magnitude and Phase





$$\theta = \mathrm{Tan}^{-1}\left(\frac{b}{a}\right)$$

Magnitude condition and Phase condition of RL



$$T(s) = \frac{Y(s)}{X(s)} = \frac{G(s)}{1+G(s)H(s)}$$

The product G(s)H(s) is a rational polynomial function can be expressed as:

$$G(s)H(s) = K\frac{(s+z_1)(s+z_2)\cdots(s+z_m)}{(s+p_1)(s+p_2)\cdots(s+p_n)}$$

1+GH=0 $\Rightarrow$ GH= -1

Magnitude condition:

A value of K satisfies the magnitude condition for a given s point of the root locus if

$$|G(s)H(s)| = 1$$

which is the same as saying that

$$K\frac{|s+z_1||s+z_2|\cdots|s+z_m|}{|s+p_1||s+p_2|\cdots|s+p_n|} = 1.$$

Phase condition (Angle condition):

A point s of the complex s–plane satisfies the angle condition if

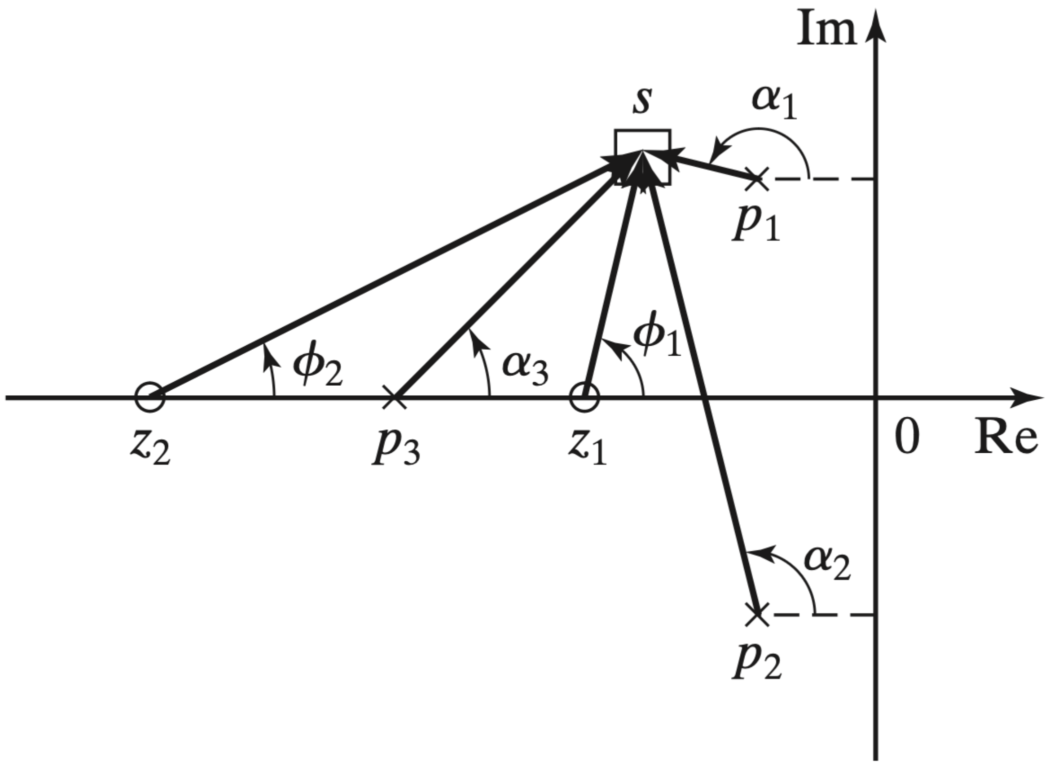
$$\angle(G(s)H(s)) = \pi$$

which is the same as saying that

$$\sum_{i=1}^{m}\angle(s+z_i) - \sum_{i=1}^{n}\angle(s+p_i) = \pi$$

Magnitude condition and Phase condition of RL

For example: $$L(s) = \frac{K(s-z_1)(s-z_2)}{(s-p_1)(s-p_2)(s-p_3)}, \qquad p_2 = \bar{p}_1.$$



Several basic rules can be derived from the phase condition, which will facilitate the sketching of the root locus, such as “asymptotes,” “breakaway points,” “angle of departure,” and “angle of arrival”.

$$\begin{aligned}\angle L(s) &= \angle(s-z_1) + \angle(s-z_2) - \angle(s-p_1) - \angle(s-p_2) - \angle(s-p_3)\\ &= \phi_1 + \phi_2 - \alpha_1 - \alpha_2 - \alpha_3.\end{aligned}$$

Proof of the "asymptotes" in RL

Consider the characteristic equation： $1+KG(s)H(s)=1+K\frac{b_0s^m+b_1s^{m-1}+\cdots+b_{m-1}s+b_m}{a_0s^n+a_1s^{n-1}+\cdots+a_{n-1}s+a_n}=0$

We can rewrite this, and then if we let |s|→∞ , all but the highest order terms of the polynomials go become insignificant.

$$KG(s)H(s)=-1$$

$$\underset{s\to\infty}{KG(s)H(s)}\approx K\frac{b_0s^m}{a_0s^n}=K\frac{b_0}{a_0}\frac{1}{s^{n-m}}=K\frac{b_0}{a_0}\frac{1}{s^q}=K\frac{b_0}{a_0}s^{-q}$$

Now we can apply the angle criterion

$$\angle KG(s)H(s)=\angle -1=\pm 180°$$

$$\angle \underset{s\to\infty}{KG(s)H(s)}=\angle K\frac{b_0}{a_0}s^{-q}=\pm r180°,\quad r=1,3,5,\ldots$$

Now we can write "s" as $s=Me^{j\theta}$ and complete the derivation

$$\angle K\frac{b_0}{a_0}\left(Me^{j\theta}\right)^{-q}=\pm r180°$$

K, a_0, b_0 and M are all positive so they don't contribute to the angle, so

$$\angle e^{-j\theta q}=\theta q=\pm r180°,\quad \text{or}$$

$$\theta=\pm r\frac{180°}{q},\quad r=1,3,5\ldots$$

https://lpsa.swarthmore.edu/Root_Locus/DeriveRootLocusRules.html

To summarize: as |s|→∞ , the loci are asymptotic to a set of lines that radiate outward from the origin with angles of θ=±r180°/q where r=1, 3, 5...

Summary of RL rules (1/2)

$\boxed{1}$. The root locus is symmetric with respect to the real axis. Why?

$\boxed{2}$. The root loci start from n poles p_i (when $K = 0$) and approach the n zeros (m finite zeros z_i and $n - m$ infinite zeros when $K \to \infty$). Rule B & C

$\boxed{3}$. The root locus includes all points on the real axis to the left of an odd number of open-loop real poles and zeros. Rule D

$\boxed{4}$. As $K \to \infty$, $n - m$ branches of the root-locus approach asymptotically $n - m$ straight lines (called **asymptotes**) with angles Rule E

$$\theta = \frac{(2k+1)180^\circ}{n-m}, \quad k = 0, \pm 1, \pm 2, \ldots$$

and the starting point of all asymptotes is on the real axis at

$$\kappa = \frac{\sum_{i=1}^{n} p_i - \sum_{j=1}^{m} z_j}{n-m} = \frac{\sum \text{poles} - \sum \text{zeros}}{n-m}.$$

Not introduced yet

Summary of RL rules (2/2)

Help us to draw the curve more precisely

Not be tested

5. The **breakaway points** (where the root loci meet and split away, usually on real axis) and the **breakin points** (where the root loci meet and enter the real axis) are among the roots of the equation: $\frac{dL(s)}{ds} = 0$. (On the real axis, only those roots that satisfy Rule 3 are breakaway or breakin points.)

6. The **departure angle** ϕ_k (from a pole, p_k) is given by

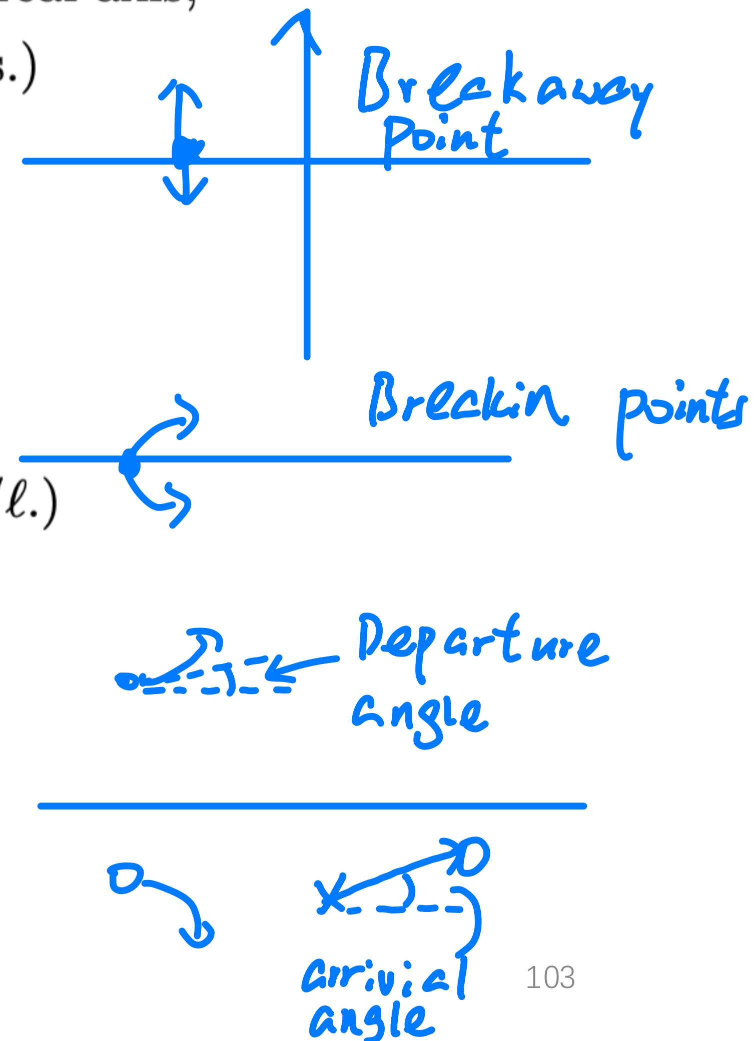
$$\phi_k = \sum_{i=1}^{m} \angle(p_k - z_i) - \sum_{j=1, j\neq k}^{n} \angle(p_k - p_j) \pm 180^\circ.$$

(In the case p_k is l repeated poles, the departure angle becomes ϕ_k/ℓ.)

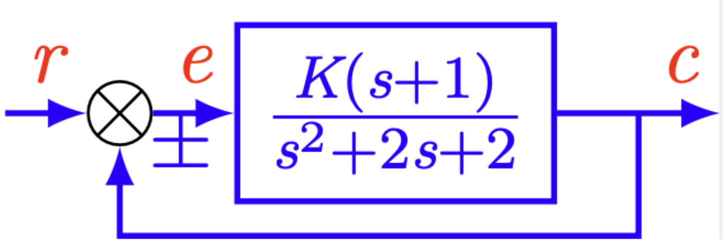
The **arrival angle** ψ_k (at a zero, z_k) is given by

$$\psi_k = -\sum_{i=1, i\neq k}^{m} \angle(z_k - z_i) + \sum_{j=1}^{n} \angle(z_k - p_j) \pm 180^\circ.$$

(In the case z_k is l repeated zeros, the arrival angle becomes ψ_k/ℓ.)



*Example of breakaway and breakin point:



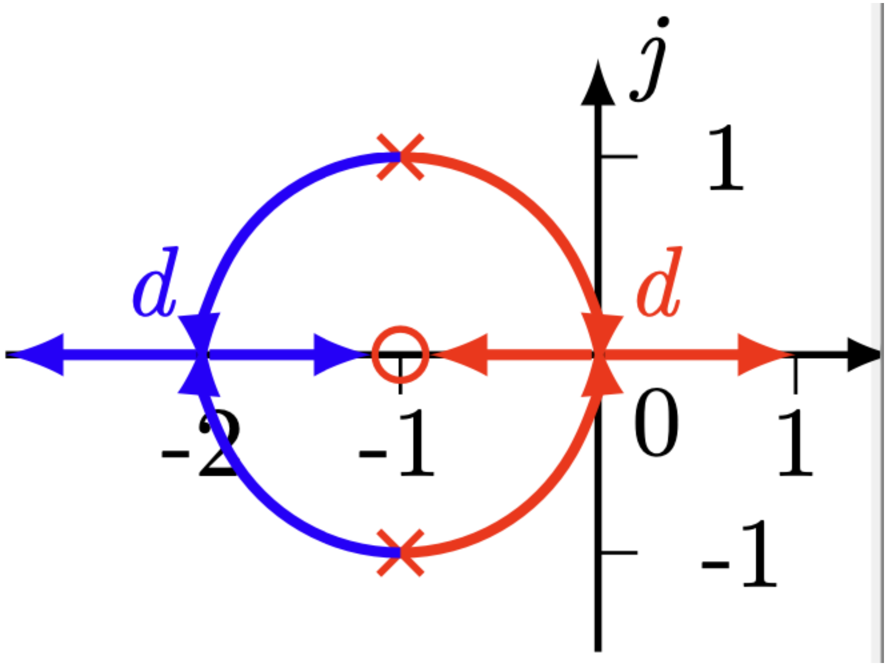
$$G(s) = \frac{K(s+1)}{s^2+2s+2} = \frac{K(s+1)}{(s+1+j)(s+1-j)}$$

$$\frac{dL(s)}{ds} = 0 \quad \Leftrightarrow \quad \sum_{i=1}^{n} \frac{1}{d-p_i} = \sum_{j=1}^{m} \frac{1}{d-z_j}$$

$$\Rightarrow \frac{1}{d+1+j} + \frac{1}{d+1-j} = \frac{2(d+1)}{d^2+2d+2} = \frac{1}{d+1}$$

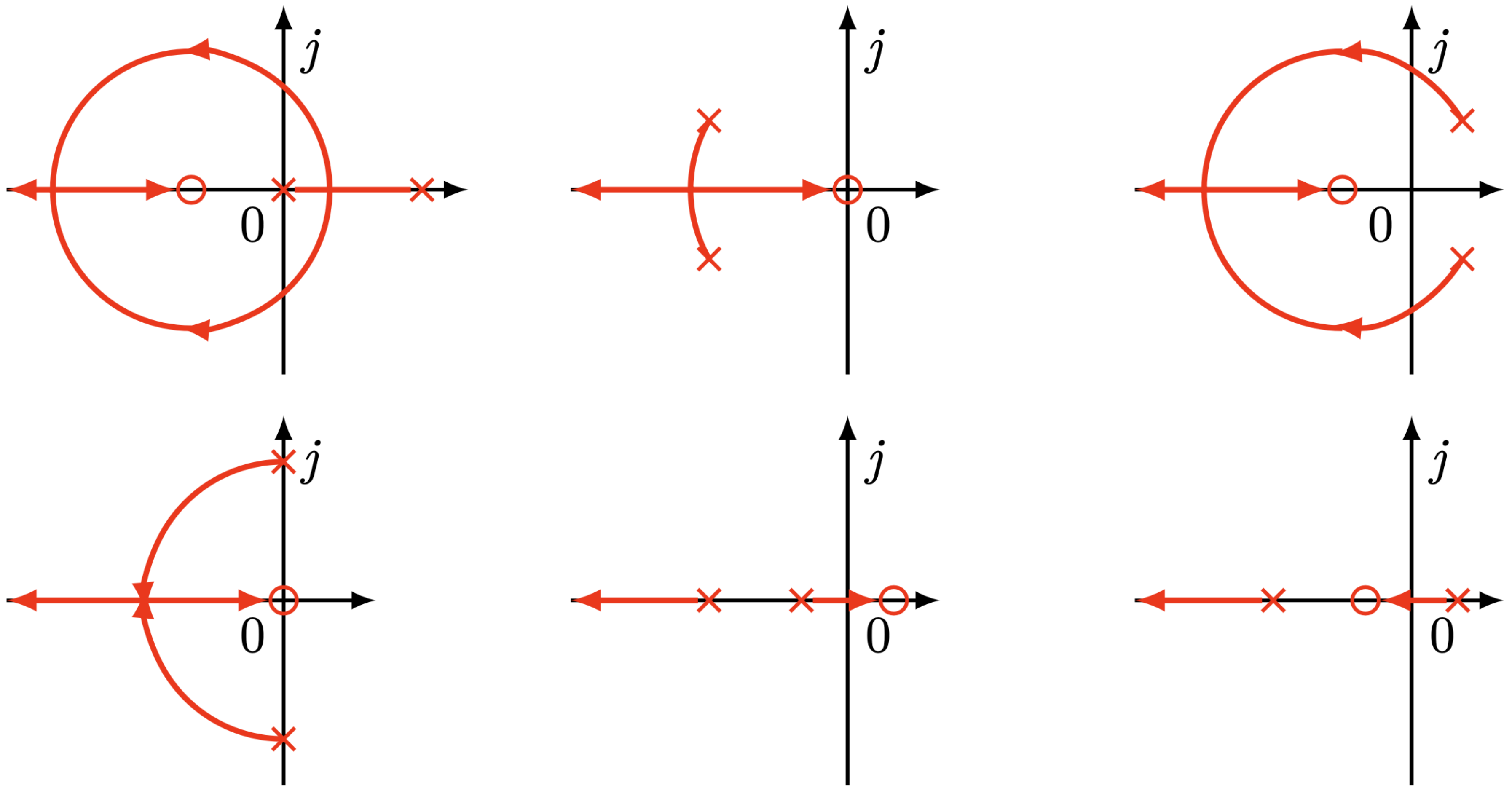
$$d^2 + 2d = d(d+2) = 0$$

$$\Rightarrow \begin{cases} d_1 = -2 \\ K_{d_1} = \frac{|d+1+||d+1-j|}{|d+1|} \overset{d=-2}{=} 2 \end{cases} \qquad \begin{cases} d_2 = 0 \\ K_{d_2} = \frac{|d+1+||d+1-j|}{|d+1|} \overset{d=0}{=} 2 \end{cases}$$



*Example of departure and arrival angles:

Theorem: For the loop transfer function with one zero and two poles, then the root locus in the complex plane must be an arc with the zero as its center.



*Example of departure and arrival angles:

In this example, departure and arrival angles are illustrated. Consider an open-loop transfer function

$$L(s) = \frac{K(s^2+4s+8)}{(s+3)(s^2+2s+2)} = \frac{K(s-z_1)(s-z_2)}{(s-p_1)(s-p_2)(s-p_3)}$$

with zeros at $z_1 = -2 + j2$ and $z_2 = -2 - j2$ and poles at $p_1 = -3$, $p_2 = -1 + j$, and $p_3 = -1 - j$.

The departure angle ϕ at $p_2 = -1 + j$ satisfies the following equation

$$\angle(p_2 - z_1) + \angle(p_2 - z_2) - \angle(p_2 - p_1) - \phi - \angle(p_2 - p_3) = -180^\circ,$$

i.e.,

$$-45^\circ + \tan^{-1} 3 - 90^\circ - \phi - \tan^{-1}\frac{1}{2} = -180^\circ$$

which gives $\phi = 90^\circ$, while the arrival angle at the zero $z_1 = -2 + j2$ can be calculated from

$$\psi + \angle(z_1 - z_2) - \angle(z_1 - p_1) - \angle(z_1 - p_2) - \angle(z_1 - p_3) = -180^\circ,$$

i.e.,

$$\psi + 90^\circ - \tan^{-1} 2 - 135^\circ - (180^\circ - \tan^{-1} 3) = -180^\circ$$

which gives $\psi = 36.87^\circ$.

The departure angle at $p_3 = -1 - j$ and the arrival angle at $z_2 = -2 - j2$ are $-\phi$ and $-\psi$, respectively. All these angles are shown in Figure 5.5.

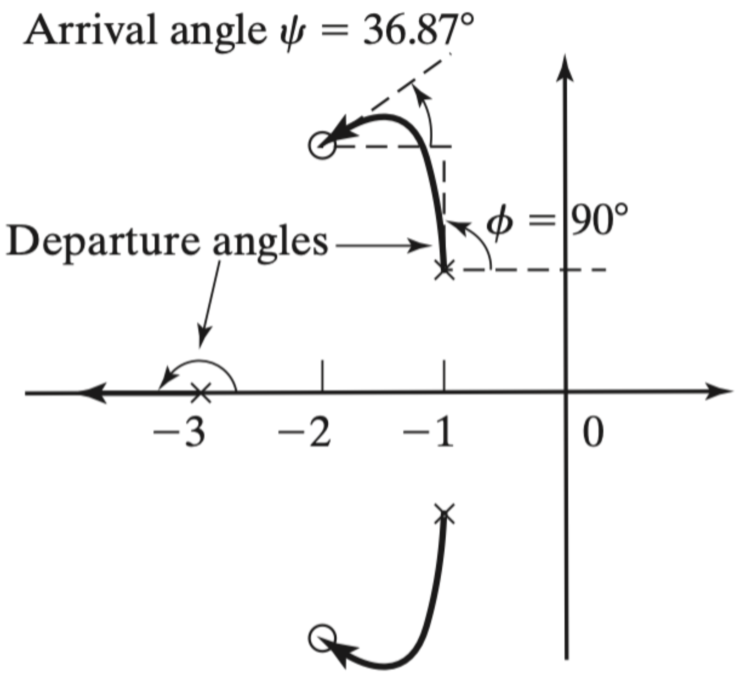
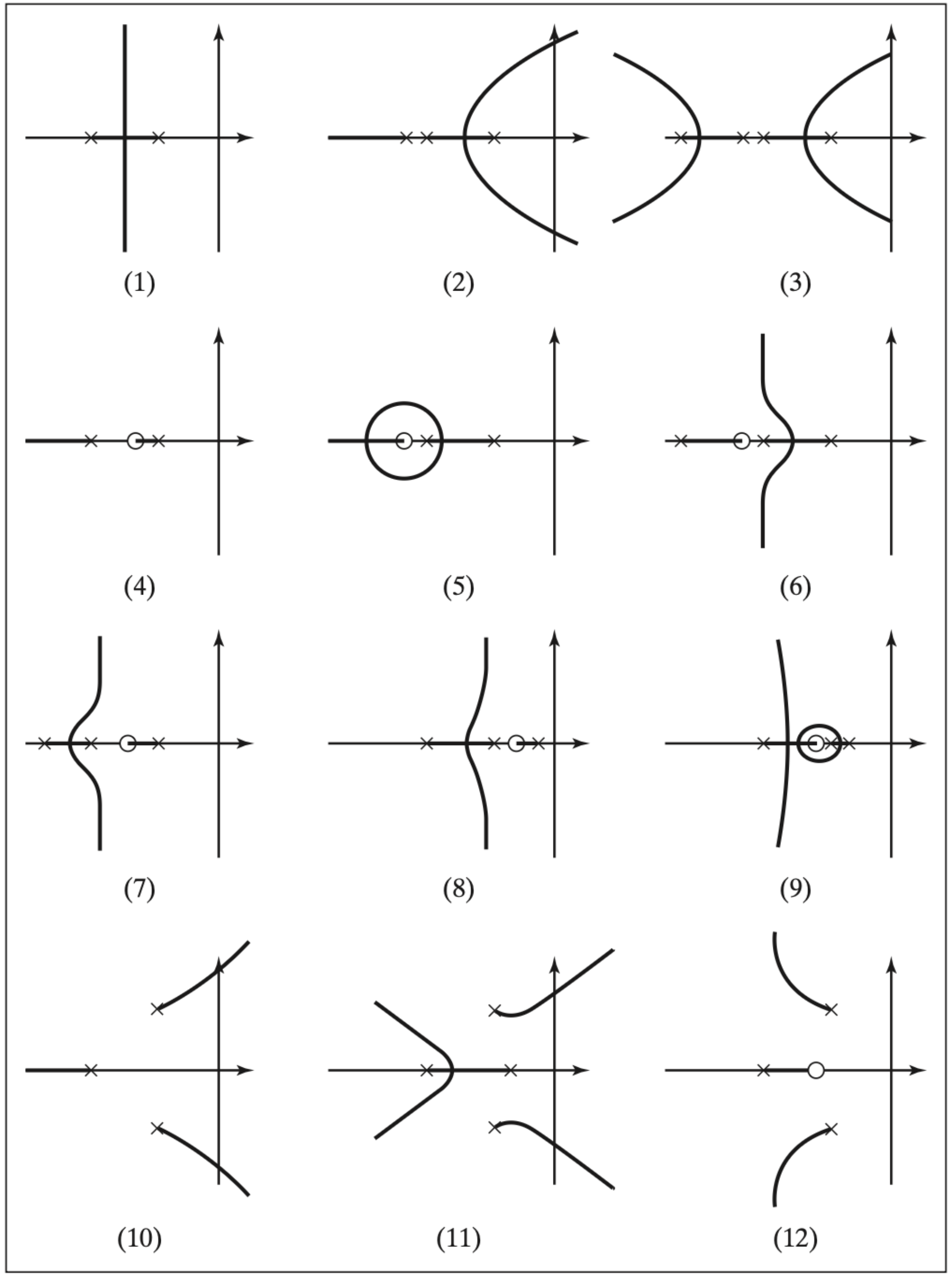


Figure 5.5: Departure and arrival angles.

For more details, please check the textbook of this course, which has been uploaded to canvas.

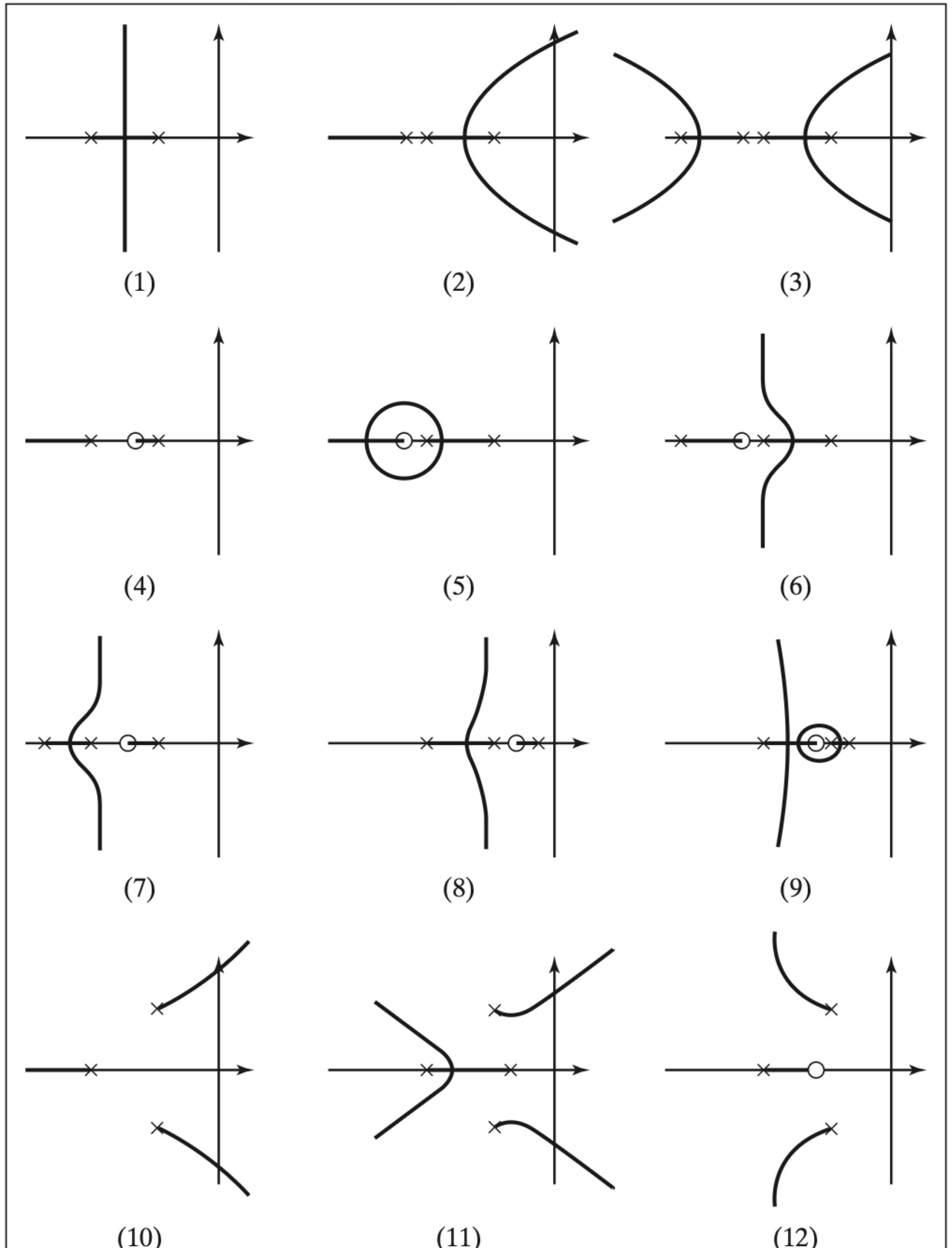
*Effects Of Adding Poles And Zeros

- **Adding open-loop poles: tend to push the root loci to the right half plane.**
  - Plot (2) can be obtained from Plot (1) by adding one more open-loop pole. It is seen that two root loci in Plot (2) go to the right half plane along two asymptotes of 60° and -60°
  - Similarly, Plot (3) can be obtained by adding two more poles to Plot (1) or one more pole to Plot (2). In this case, two root loci in Plot (3) go to the right half plane along two asymptotes of 45° and -45°, meaning these two root loci will go to the right half plane faster than the two in Plot (2) as the parameter is increased.
  - The same conclusions can be drawn by comparing Plot (10) with Plot (11).



*Effects Of Adding Poles And Zeros

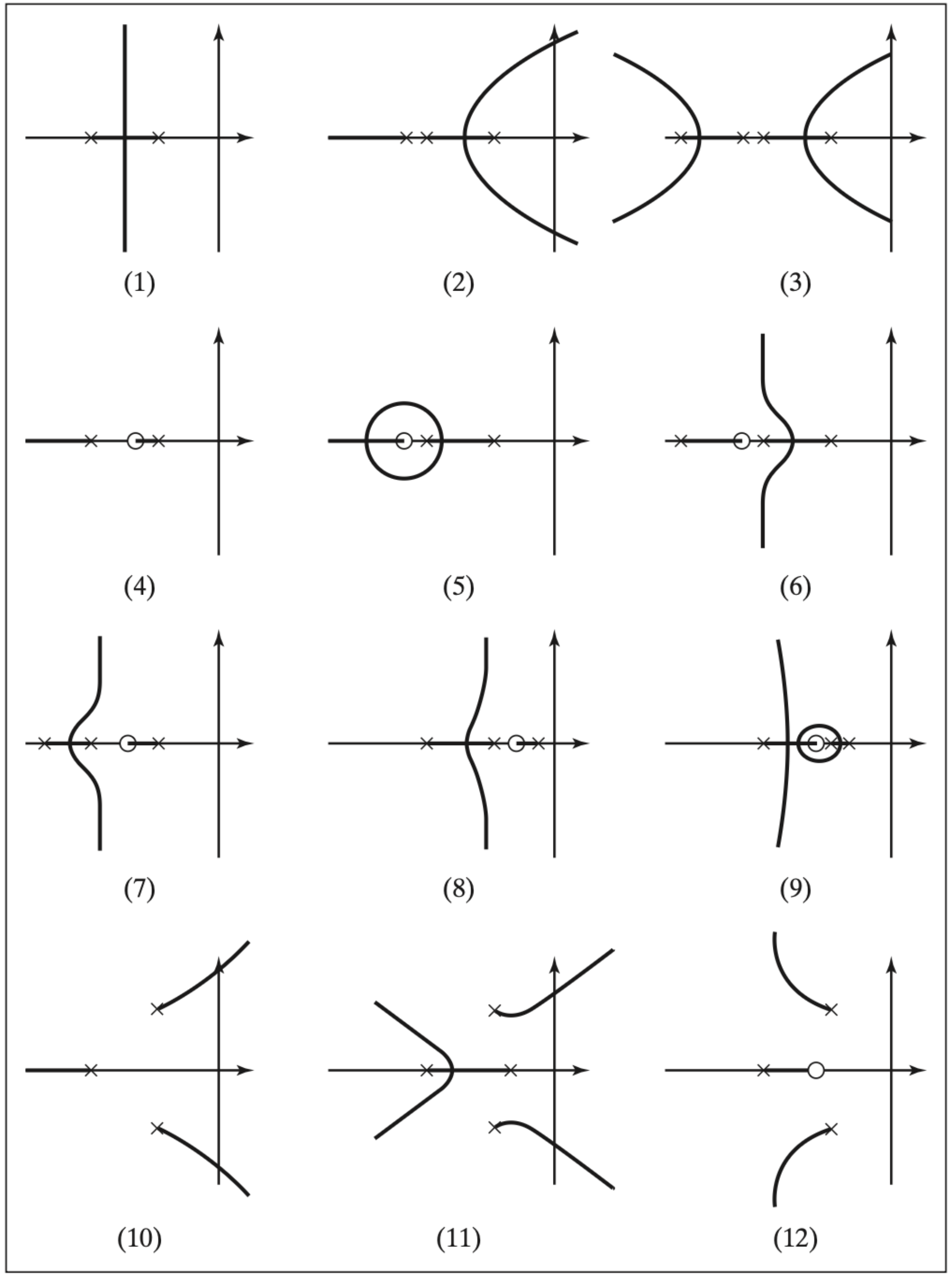
- **Adding open-loop zeros (PD controllers): Zeros attract root loci, and thus suitably placed zeros can move root loci to the desired region.**
  - Compare Plot (1) with Plot (4) and Plot (5). Indeed, Plot (4) and Plot (5) can be obtained by adding one zero to Plot (1).
  - It is not hard to see that the additional zero tends to attract the root loci. Thus, by suitably placing the additional zero, one can move the root loci to the desired region. This is how the PD controller works.
  - Similar conclusions can be reached by comparing Plot (10) with Plot (12).



*Effects Of Adding Poles And Zeros

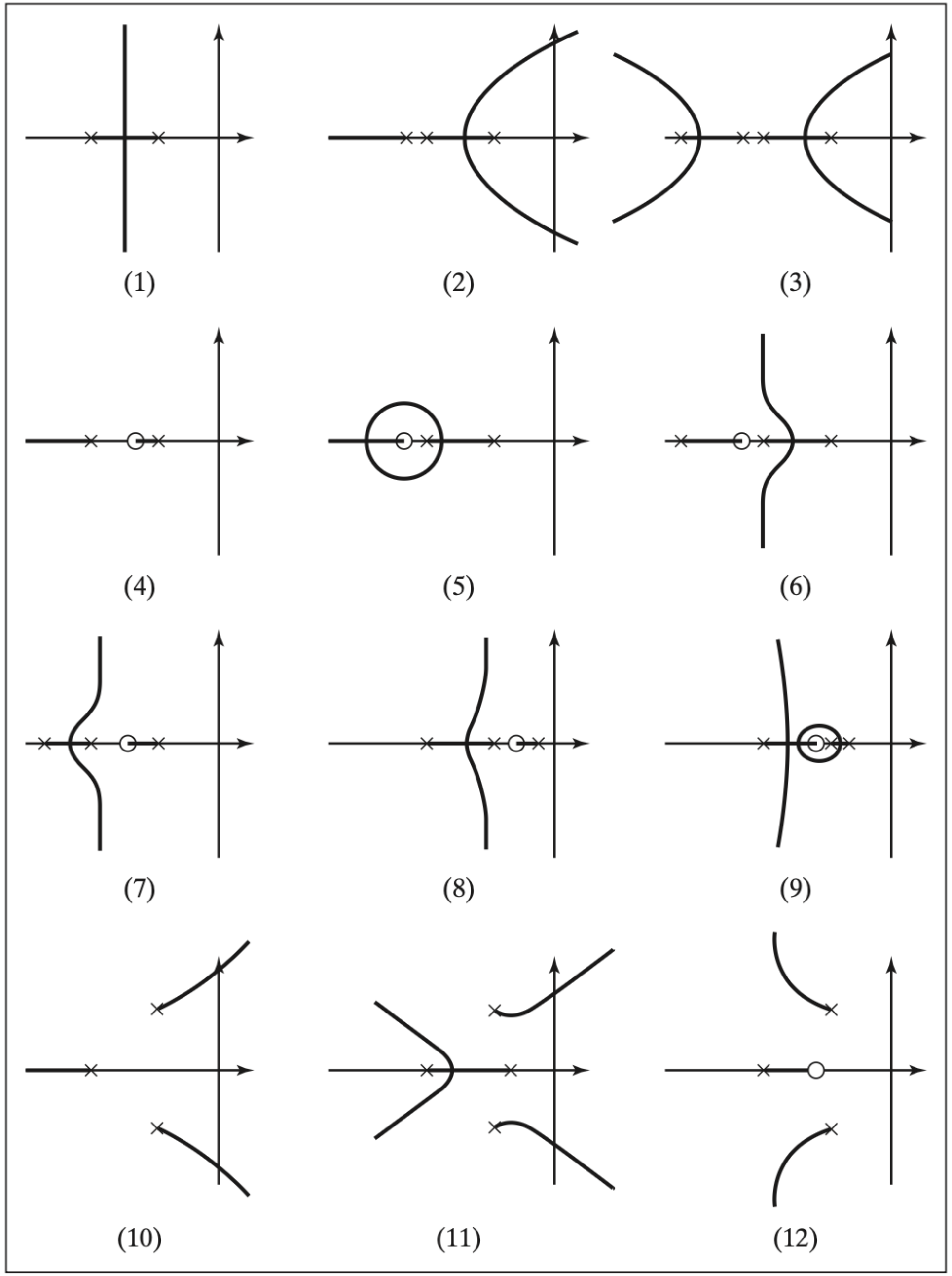
- Adding a pair of pole and zero with zero on the right of the pole (Lead controllers): A lead (or PD) controller can be used to improve the transient performance if it is suitably designed.

  - Compare Plot (1) with Plot (6) and Plot (7). We can think that Plot (6) and Plot (7) are obtained by adding a pair of a pole and a zero to Plot (1) with the zero on the right of the pole.
  - In both cases, two root loci tend to move left toward the stable region while another root-locus approaches the open-loop zero.
  - When a closed-loop pole is close enough to a zero, their effects on the closed-loop performance become negligible since this closed-loop pole will almost be canceled by this zero. Thus, the other two root loci play the dominant roles.
  - Since these two root loci are in the more stable region, it is expected that the closed-loop performance with this lead (or PD) controller will be improved when the controller parameters are suitably chosen.



*Effects Of Adding Poles And Zeros

- Adding a pair of a pole and a zero with a pole on the right of the zero (Lag controllers): A lag (or PI) controller can be used to improve steady-state tracking.
    - Compare Plot (1) with Plot (8) and Plot (9). Plot (8) and Plot (9) are obtained from Plot (1) by adding a pair of a pole and a zero with the pole on the right of the zero.
    - It is clear that the total effect is somewhat similar to adding a pole. Thus, part of the root-locus plot is pushed to the right to a certain degree.
    - It is also important to note that the root-locus plot will not change much (except the local part near the pair of a pole and a zero) if the pair of a pole and a zero are much closer than other poles and zeros. This is indeed how a lag or PI controller works.
    - By choosing the pair of a pole and a zero much closer to one another (usually also close to origin), this controller will essentially not change the dominant pole locations. Thus, it will not change the transient response much if it is suitably designed, but will improve steady-state tracking by increasing the system gain with the ratio of zero over the pole. (This is why the pole and zero are sometimes chosen to be close to the origin so that a large ratio can be obtained.)



This example is from textbook: Chapter 5, Page 204

*An example of phase-lag controller design based on Root Locus

EXAMPLE 5.6

Consider a unity feedback system with a plant model

$$P(s) = \frac{10}{s(s+5)(s+10)}.$$

We would like to design a controller so that

- the response of the closed-loop system with respect to a step input has no more than 20% overshoot and the settling time is no greater than 4 [sec];
- the steady-state error with respect to a ramp input is no more than 0.05.

Solution:

First of all, we shall find the desired dominant pole locations. Suppose that the dominant poles are a pair of complex poles, s_1 and $\bar{s}_1$, given as the poles of a standard second-order system

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2},$$

i.e.,

$$s_1 = -\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2}.$$

Then, to guarantee that the overshoot is no more than 20%, we need the damping ratio to satisfy

$$\zeta \geq \frac{-\ln(20/100)}{\sqrt{\pi^2 + (\ln 20/100)^2}} = 0.456.$$

To guarantee that the settling time is no more than 4 [sec] with 2% tolerance, we need $\zeta\omega_n \geq 4/t_s = 1$. We shall pick s_1 in the region described above in the following design process.

We shall now follow the phase-lag design procedure to see whether a phase-lag controller can be designed to satisfy the above design specifications.

Solution (continue⋯):

- We shall first construct a root-locus plot for

$$KP(s) = \frac{10K}{s(s+5)(s+10)}.$$

```
P=zpk([],[0,-5,-10],10);

rlocus(P);

sgrid      draw damping ratio
    lines
```

Then point and click on the desired point to find K_0 and s_1

```
[K0,s1]=rlocfind(P)
```

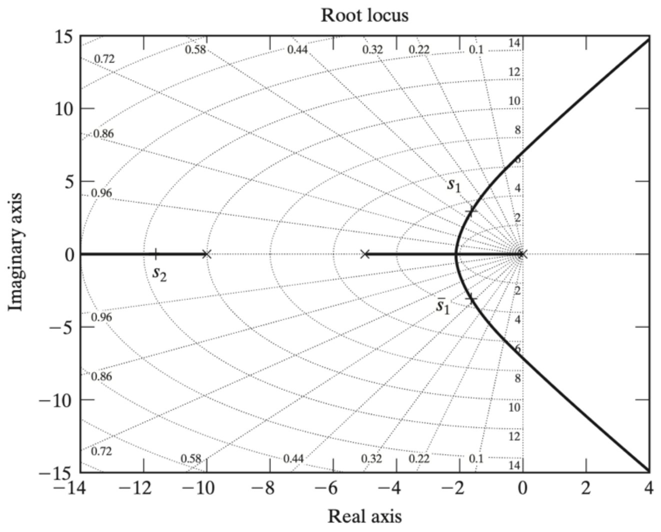
The root locus is shown in Figure 5.10. Now, choose the dominant poles on the root-locus plot so that their real parts are greater than 1 and they have approximately the damping ratio 0.5 by using the following MATLAB command

```
>> [K0,s1]=rlocfind(P)
```

We then get

$$K_0 = 13,\ s_1 = -1.665 + j2.8927,$$

$$\bar{s}_1 = -1.665 - j2.8927,\ s_2 = -11.67.$$



Solution (continue⋯):

- We shall now find the required gain to satisfy the desired steady-state error. Note that $K_s = C(0)$ and

$$K_v = \lim_{s\to 0} sP(s)C(s) = \frac{C(0)}{5}.$$

Thus

$$e_{ss} = \frac{1}{K_v} = \frac{5}{C(0)} \leq 0.05$$

gives $K_s = C(0) \geq 100$. Take $K_s = 100$.

- Take $b = 0.05$, which is much smaller than $|s_1|$. Then $a = K_0 b/K_s = 0.0065$ and we have a controller

```
b=0.05;   K0=13;   Ks=100;
a=K0*b/Ks;
C=zpk(-b,-a,K0);  controller
T=feedback(P*C,1);  closed-loop
pole(T)  closed-loop poles
step(T) step response
```

$$C(s) = \frac{13(s+0.05)}{s+0.0065}$$

which gives the closed-loop poles at

$$-11.6656,\ -0.0509,$$

$$-1.6450 \pm j2.8724$$

and a closed-loop zero at -0.05, which approximately cancels the closed-loop pole at -0.0509.

Solution (continue⋯):

Thus, we can see that the closed-loop system has two dominant poles at $-1.6450 \pm j2.8724$, which are very close to s_1 and $\bar{s}_1$ as expected. The step response and the ramp response are shown in Figures 5.11 and 5.12, respectively. The design specifications are met with a settling time of 1.89 [sec] and an overshoot of 17.7%.

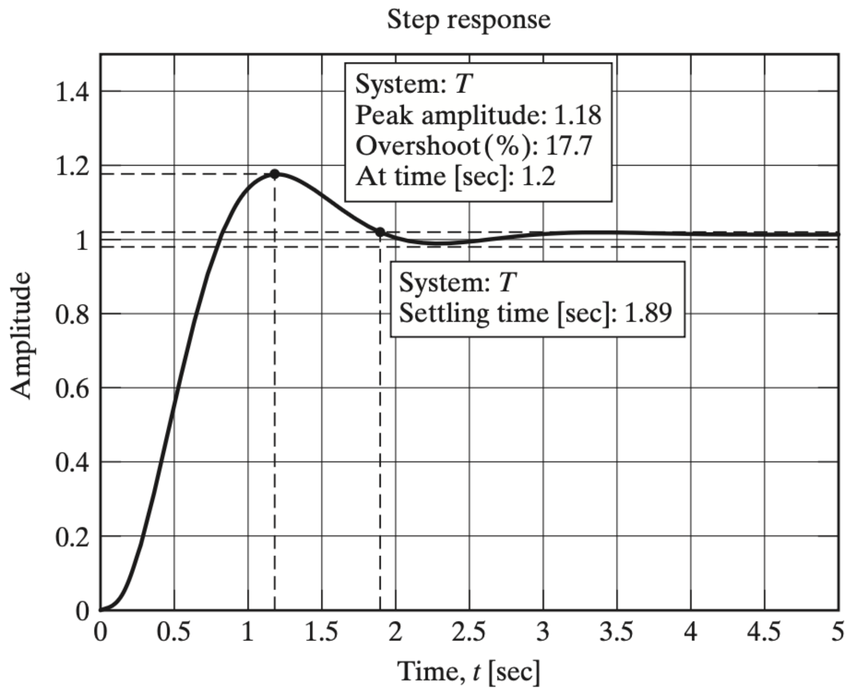


Figure 5.11: Step response of Example 5.6.

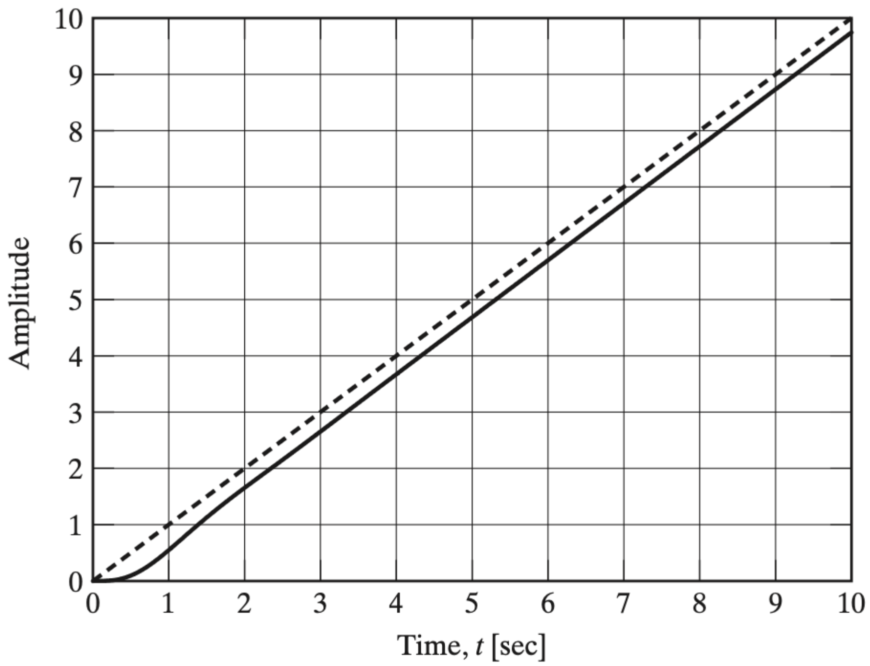


Figure 5.12: Ramp response of Example 5.6.